# Chapter 14

*Campbell Biology, AP Edition, 12e*

Flowers are a pinkish-purple.

# Chapter 14

# Mendel and the Gene Idea

**Figure 14.1**

Flower color was one of the many characteristics of pea plants studied by Gregor Mendel. By breeding pea plants over many generations and carefully counting the types of offspring, Mendel developed a theory of inheritance that is the basis of modern genetics.

How are traits, such as the purple or white color of flowers, transmitted from parents to offspring? Alternative versions of a gene (alleles) account for different traits.

Each parent cell has two alleles for each character, such as flower color. Here, the alleles are identical in each parent flower. The two alleles segregate (separate) during gamete formation, ending up in different gametes (Law of Segregation). Offspring inherit one allele from each parent. When the two alleles are different, one allele (here, purple) determines the appearance. When offspring reproduce, the same principle applies: The alleles segregate, ending up in different gametes.

## Key Concepts

- 14.1**Mendel used the scientific approach to identify two laws of inheritance**
- 14.2**Probability laws govern Mendelian inheritance**
- 14.3**Inheritance patterns are often more complex than predicted by simple Mendelian genetics**
- 14.4**Many human traits follow Mendelian patterns of inheritance**

Advanced placement edition

The meticulous work of Mendel and others in the 19th century provided evidence for a particulate mechanism of inheritance that led to our current understanding of the chromosomal basis of inheritance of

genes from parent to offspring **(BIG IDEA 3)** and provided structural and functional evidence to support the relatedness of all domains **(BIG IDEA 1)**.

## Study Tip

**Solving genetics problems:** In Figure 14.5, you'll learn how to predict the offspring of a genetic cross with a Punnett square, like the example shown below. For crosses shown in other figures, cover up the Punnett square and draw your own. Also, check out the Tips for Genetics Problems at the end of the chapter.

A Punnett square has rows for Eggs: uppercase P and lowercase p. It has columns for sperm: uppercase P and lowercase p.

A Punnett square illustrates the inheritance of alleles uppercase P and lowercase p, by depicting a square divided into 4 equal square sections. There are 2 rows and 2 columns forming 4 cells. To the left of the two rows is labeled Eggs, where the gamete with uppercase P (encircled) is placed near row 1 and the gamete with lowercase p (encircled) is placed near row 2. Above the columns is labeled sperm where the gamete with uppercase P (encircled) is placed near column 1 and the gamete with uppercase P (encircled) is placed near column 2. Top left quadrant is labeled by uppercase P uppercase P, top right quadrant is labeled uppercase p lowercase p, bottom left quadrant is labeled uppercase P lowercase p, and bottom right quadrant is labeled lowercase p lowercase p.

## Get Ready for This Chapter

In this chapter, you will learn about Mendelian genetics and more complex inheritance patterns, and will explore the rules of probability and the application of Mendelian principles to human genetics. Before beginning this chapter, you should be able to do the following:

- Describe the chromosomal makeup of a diploid cell (see Concept 13.2 and Figure 13.4).
- Describe how a gene codes for a protein (see Concept 5.5 and Figure 5.22).
- Explain how a protein's structure determines its function, using sickle-cell disease as an example (see Concept 5.4, Figure 5.18, and Figure 5.19).

The following questions provide a quick check of your basic knowledge in these areas.

1. What number and types of chromosomes are found in a somatic cell in an animal with a diploid number of 48?

24 chromosomes ( 12 from each parent, including one sex chromosome from each)

That's incorrect. Try again.

**Incorrect.** Review the following material: All somatic cells in animals have two sets of chromosomes, one haploid set (n) from each parent, and each set includes one sex chromosome. For an animal with a diploid number of 48 2n=48, the number of chromosomes in the somatic cells is 48 and the number of chromosomes in each haploid set is 24 n=24. Humans have a diploid number of 46 2n=46, and the number of chromosomes in each set is 23 2n=46, including one sex chromosome.

96 chromosomes ( 48 from each parent, including one pair of sex chromosomes from each)

That's incorrect. Try again.

**Incorrect.** Review the following material: All somatic cells in animals have two sets of chromosomes, one haploid set (n) from each parent, and each set includes one sex chromosome. For an animal with a diploid number of 48 2n=48, the number of chromosomes in the somatic cells is 48 and the number of chromosomes in each haploid set is 24 (n=24). Humans have a diploid number of 46 (2n=46), and the number of chromosomes in each set is 23 (n=23), including one sex chromosome.

23 chromosomes ( 11 from each parent, plus one sex chromosome)

That's incorrect. Try again.

Review the following material: All somatic cells in animals have two sets of chromosomes, one haploid set (n) from each parent, and each set includes one sex chromosome. For an animal with a diploid number of 48 (2n = 48), the number of chromosomes in the somatic cells is 48 and the number of chromosomes in each haploid set is 24 (n=24). Humans have a diploid number of 46 (2n=46), and the number of chromosomes in each set is 23 (n=23), including one sex chromosome.

48 chromosomes ( 24 from each parent, including one sex chromosome from each)

That's correct.

**Correct!**If you had trouble with this question, review the following material: All somatic cells in animals have two sets of chromosomes, one haploid set (n) from each parent, and each set includes one sex chromosome. For an animal with a diploid number of 48 (2n=48), the number of chromosomes in the somatic cells is 48 and the number of chromosomes in each haploid set is 24 (n=24). Humans have a diploid number of 46 (2n=46), and the number of chromosomes in each set is 23 (n=23), including one sex chromosome.

2. The information contained in DNA is used to make which of the following product(s)?

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tRNA only
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That's incorrect. Try again.

**Incorrect.** Review the following material: The information that programs all of a cell's activities is encoded in the sequence of DNA nucleotides. DNA directs RNA synthesis and, through mRNA, directs the order of the amino acids during protein synthesis; this entire process is called gene expression. Two types of RNA required for protein synthesis are mRNA and tRNA. (Other RNAs, such as rRNAs, are also necessary, and these too are encoded by genes in DNA.)

```
proteins only
```

That's incorrect. Try again.

**Incorrect.** Review the following material: The information that programs all of a cell's activities is encoded in the sequence of DNA nucleotides. DNA directs RNA synthesis and, through mRNA, directs the order of the amino acids during protein synthesis; this entire process is called gene expression. Two types of RNA required for protein synthesis are mRNA and tRNA. (Other RNAs, such as rRNAs, are also necessary, and these too are encoded by genes in DNA.)

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mRNA only
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That's incorrect. Try again.

**Incorrect.** Review the following material: The information that programs all of a cell's activities is encoded in the sequence of DNA nucleotides. DNA directs RNA synthesis and, through mRNA, directs the order of the amino acids during protein synthesis; this entire process is called gene expression. Two types of RNA required for protein synthesis are mRNA and tRNA. (Other RNAs, such as rRNAs, are also necessary, and these too are encoded by genes in DNA.)

```
proteins, mRNA, and tRNA
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That's correct.

**Correct!** If you had trouble with this question, review the following material: The information that programs all of a cell's activities is encoded in the sequence of DNA nucleotides. DNA directs RNA synthesis and, through mRNA, directs the order of the amino acids during protein synthesis; this entire process is called gene expression. Two types of RNA required for protein synthesis are mRNA and tRNA. (Other RNAs, such as rRNAs, are also necessary, and these too are encoded by genes in DNA.)

3. Normal hemoglobin is a tetramer, consisting of two molecules of beta-globin and two molecules of alpha-globin; normal hemoglobin molecules do not associate with each other. In sickle-cell disease, the change in a single amino acid results in mutant hemoglobin tetramers, which associate with each other and assemble into large fibers. Based on this information alone, what can we conclude about the changes in the structure of sickle cell hemoglobin as compared to normal hemoglobin?

```
altered primary and quaternary structure
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That's correct.

**Correct!** If you had trouble with this question, review the following material: The primary structure of a protein is its sequence of amino acids, secondary structure is the result of hydrogen bonds between the repeating constituents of the polypeptide backbone, tertiary structure is the three-dimensional shape stabilized by interactions between side chains, and quaternary structure results when two or more polypeptides associate with each other. Normal hemoglobin tetramers do not associate with each other. The single amino acid change that causes sickle-cell disease alters the primary and quaternary structure of the protein, and as a result, the abnormal hemoglobin tetramers aggregate into chains. Not enough information is given to determine whether secondary and tertiary interactions are affected.

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altered primary structure only
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That's incorrect. Try again.

**Incorrect.** Review the following material: The primary structure of a protein is its sequence of amino acids, secondary structure is the result of hydrogen bonds between the repeating constituents of the polypeptide backbone, tertiary structure is the three-dimensional shape stabilized by interactions between side chains, and quaternary structure results when two or more polypeptides associate with each other. Normal hemoglobin tetramers do not associate with each other. The single amino acid change that causes sickle-cell disease alters the primary and quaternary structure of the protein, and as a result, the abnormal hemoglobin tetramers aggregate into chains. Not enough information is given to determine whether secondary and tertiary interactions are affected.

```
altered secondary and tertiary structure
```

That's incorrect. Try again.

**Incorrect.** Review the following material: The primary structure of a protein is its sequence of amino acids, secondary structure is the result of hydrogen bonds between the repeating constituents of the polypeptide

backbone, tertiary structure is the three-dimensional shape stabilized by interactions between side chains, and quaternary structure results when two or more polypeptides associate with each other. Normal hemoglobin tetramers do not associate with each other. The single amino acid change that causes sickle-cell disease alters the primary and quaternary structure of the protein, and as a result, the abnormal hemoglobin tetramers aggregate into chains. Not enough information is given to determine whether secondary and tertiary interactions are affected.

```
altered quaternary structure only
```

That's incorrect. Try again.

**Incorrect.** Review the following material: The primary structure of a protein is its sequence of amino acids, secondary structure is the result of hydrogen bonds between the repeating constituents of the polypeptide backbone, tertiary structure is the three-dimensional shape stabilized by interactions between side chains, and quaternary structure results when two or more polypeptides associate with each other. Normal hemoglobin tetramers do not associate with each other. The single amino acid change that causes sickle-cell disease alters the primary and quaternary structure of the protein, and as a result, the abnormal hemoglobin tetramers aggregate into chains. Not enough information is given to determine whether secondary and tertiary interactions are affected.

## Learning Objectives

After studying this chapter, you should be able to do the following:

- 14.1Explain Mendel's two laws of inheritance.
- 14.2Describe the laws of probability and explain how they can be used to solve complex genetics problems.
- 14.3Use examples to show how genetic inheritance patterns can be affected by complete dominance, incomplete dominance, codominance, multiple alleles, pleiotropy, epistasis, and polygenic inheritance.
- 14.4Explain what a pedigree is and use examples to show how human traits follow Mendelian patterns of inheritance.

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# Concept 14.1: Mendel used the scientific approach to identify two laws of inheritance

Modern genetics began during the mid-1800s with a monk named Gregor Mendel, who discovered the basic principles of heredity by breeding garden peas in carefully planned experiments. As we retrace his work, you will recognize the key elements of the scientific process that were introduced in Concept 1.3.

## Mendel's Experimental, Quantitative Approach

Mendel grew up on his parents' small farm in a region of Austria that is now part of the Czech Republic. In this agricultural area, Mendel received agricultural training in school along with a basic education. As an adolescent, Mendel overcame financial hardship and illness to excel in high school and, later, at the Olmutz Philosophical Institute.

In 1843, at the age of 21, Mendel entered an Augustinian monastery, a reasonable choice at that time for someone who valued the life of the mind. He considered becoming a teacher but failed the necessary examination. In 1851, he left the monastery to pursue two years of study in physics and chemistry at the University of Vienna. These were very important years for Mendel's development as a scientist, in large part due to the strong influence of two professors. One was the physicist Christian Doppler, who encouraged his students to learn science through experimentation and trained Mendel to use mathematics to help explain natural phenomena. The other was a botanist named Franz Unger, who aroused Mendel's interest in the causes of variation in plants.

After attending university, Mendel returned to the monastery and was assigned to teach at a local school, where several other instructors were enthusiastic about scientific research. In addition, his fellow monks shared a long-standing fascination with the breeding of plants. Around 1857, Mendel began breeding garden peas in the abbey garden to study inheritance. Although the question of heredity had long been a focus of curiosity at the monastery, Mendel's fresh approach allowed him to deduce principles that had remained elusive to others.

One reason Mendel probably chose to work with peas is that there are many varieties. For example, one variety has purple flowers, while another variety has white flowers. A heritable feature that varies among individuals, such as flower color, is called a **character**. Each variant for a character, such as purple or white color for flowers, is called a **trait**.

Other advantages of using peas are their short generation time and the large number of offspring from each mating. Furthermore, Mendel could strictly control mating between plants **(Figure 14.2)**. Each pea flower has both pollen-producing organs (stamens) and an egg-bearing organ (carpel). In nature, pea plants usually

self-fertilize: Pollen grains from the stamens land on the carpel of the same flower, and sperm released from the pollen grains fertilize eggs present in the carpel.* To achieve cross-pollination of two plants, Mendel removed the immature stamens of a plant before they produced pollen and then dusted pollen from another plant onto the altered flowers (see Figure 14.2). Each resulting zygote then developed into a plant embryo encased in a seed (a pea). His method allowed Mendel to always be sure of the parentage of new seeds.

# Figure 14.2

# Research Method: Crossing Pea Plants

## Application

By crossing, or mating, two true-breeding varieties of an organism, scientists can study patterns of inheritance. In this example, Mendel crossed pea plants that varied in flower color.

## Technique

A diagram of crossing pea plants.

One. Removed stamens from a purple flower.

Two. Transferred sperm-bearing pollen from the stamens of a white flower to the egg-bearing carpel of a purple flower.

Three. Waited for pollinated carpel to mature into the pod.

Four. Planted seeds from pod.

## Results

When pollen from a white flower was transferred to a purple flower, the first-generation hybrids all had purple flowers. The result was the same for the reciprocal cross, which involved the transfer of pollen from purple flowers to white flowers.

The results of the pea plant experiment. Five. First filial generation offspring, F 1. Examined offspring: all had purple flowers.

Mendel chose to track only those characters that occurred in two distinct, alternative forms, such as purple or white flower color. He also made sure that he started his experiments with varieties that were **true-breeding**—that is, over many generations of self-pollination, these plants had produced only the same

variety as the parent plant. For example, a plant with purple flowers is true-breeding if the seeds produced by self-pollination in successive generations all give rise to plants that also have purple flowers.

In a typical breeding experiment, Mendel cross-pollinated two contrasting, true-breeding pea varieties—for example, purple-flowered plants and white-flowered plants (see Figure 14.2). This mating, or *crossing*, of two true-breeding varieties is called **hybridization**. The true-breeding parents are referred to as the **P generation** ****(parental generation), and their hybrid offspring are the ****F1 generation** (first filial generation, the word *filial* from the Latin word for "son"). Allowing these F1 hybrids to self-pollinate (or to cross-pollinate with other F1 hybrids) produces an **F2 generation** (second filial generation). Mendel usually followed traits for at least the P, F1, and F2 generations. Had he stopped his experiments with the F1 generation, the basic patterns of inheritance would have eluded him. Mendel's quantitative analysis of the F2 plants from thousands of genetic crosses like these allowed him to deduce two fundamental principles of heredity, now called the law of segregation and the law of independent assortment.

## The Law of Segregation

The explanation of heredity most widely in favor during the 1800s was the "blending" hypothesis, the idea that genetic material contributed by the two parents mixes just as blue and yellow paints blend to make green. This hypothesis predicts that over many generations, a freely mating population will give rise to a uniform population of individuals, something we don't see.

If the blending model were correct, the F1 hybrids from Mendel's cross between purple-flowered and white-flowered pea plants would have been pale purple flowers, a trait intermediate between those of the P generation. Notice in Figure 14.2 that the experiment produced a very different result: All the F1 offspring had flowers of the same color as the purple-flowered parents. What happened to the white-flowered plants' genetic contribution to the hybrids? If it were lost, then the F1 plants could produce only purple-flowered offspring in the F2 generation. But when Mendel allowed the F1 plants to self- or cross-pollinate and planted their seeds, the white-flower trait reappeared in the F2 generation. The blending hypothesis is also inconsistent with this reappearance of traits after they've skipped a generation.

Mendel used very large sample sizes and kept accurate records of his results: 705 of the F2 plants had purple flowers, and 224 had white flowers. These data fit a ratio of approximately three purple to one white **(Figure 14.3)**. Mendel reasoned that the heritable factor for white flowers did not disappear in the F1 plants but was somehow hidden, or masked, when the purple-flower factor was present. In Mendel's terminology, purple flower color is a *dominant* trait, and white flower color is a *recessive* trait. The reappearance of white-flowered plants in the F2 generation was evidence that the heritable factor causing white flowers had not been diluted or destroyed by coexisting with the purple-flower factor in the F1 hybrids. Mendel observed the same pattern of inheritance in six other characters, each represented by two distinctly different traits **(Table 14.1)**. For example, when Mendel crossed a true-breeding variety that produced

smooth, round pea seeds with one that produced wrinkled seeds, all the F1 hybrids produced round seeds; this is the dominant trait for seed shape. In the F2 generation, approximately 75% of the seeds were round and 25% were wrinkled—a 3:1 ratio, as in Figure 14.3. Now let's see how Mendel deduced the law of segregation from his experimental results. In the discussion that follows, we will use modern terms instead of some of the terms used by Mendel. (For example, we'll use "gene" instead of Mendel's "heritable factor.")

## Figure 14.3

## Inquiry: When F1 hybrid pea plants self- or cross-pollinate, which traits appear in the F2 generation?

### Experiment

Mendel crossed true-breeding purple-flowered plants and white-flowered plants (crosses are symbolized by ×). The resulting F1 hybrids were allowed to self-pollinate or were cross-pollinated with other F1 hybrids. The F2 generation plants were then observed for flower color.

P generation (true-breeding parents): purple flowers, crossed with, white flowers. F 1 generation (hybrids): all plants had purple flowers; self- or cross-pollination. F 2 generation: 705 purple-flowered plants, 224 white-flowered plants.

### Results

Both purple-flowered and white-flowered plants appeared in the F2 generation, in a ratio of approximately 3:1.

### Conclusion

The "heritable factor" for the recessive trait (white flowers) had not been destroyed, deleted, or "blended" in the F1 generation but was merely masked by the presence of the factor for purple flowers, which is the dominant trait.

**Data from** G. Mendel, Experiments in plant hybridization, *Proceedings of the Natural History Society of Brünn* 4:3–47 (1866).

**WHAT IF?** *If you mated two purple-flowered plants from the P generation, what ratio of traits would you expect to observe in the offspring? Explain. What might Mendel have concluded if he stopped his experiment after the* F1 *generation?*

*For suggested answer, see*Appendix A.**

**Table 14.1**

The Results of Mendel's F1 Crosses for Seven Characters in Pea Plants

Table 14.1 gives the results of Mendel's F 1 crosses for seven characters in pea plants.

The table is as follows:

| Character | Dominant trait | Recessive trait | F2 generation dominant: recessive | Ratio |
|---|---|---|---|---|
| flower color | purple | white | 705 to 224 | 3.15 to 1 |
| seed color | yellow | green | 6,022 to 2,001 | 3.01 to 1 |
| seed shape | round | wrinkled | 5,474 to 1,850 | 2.96 to 1 |
| pod color | green | yellow | 428 to 152 | 2.82 to 1 |
| pod shape | inflated | constricted | 882 to 299 | 2.95 to 1 |
| flower position | axial | terminal | 651 to 207 | 3.14 to 1 |
| stem length | tall | dwarf | 787 to 277 | 2.84 to 1 |

## Mendel's Model

Mendel developed a model to explain the 3:1 inheritance pattern that he consistently observed among the F2 offspring in his pea experiments. We'll now explore four related concepts making up Mendel's model, the fourth of which is the law of segregation.

First, *alternative versions of genes account for variations in inherited characters*. The gene for flower color in pea plants, for example, exists in two versions, one for purple flowers and the other for white flowers. These alternative versions of a gene are called **alleles**. Today, we can relate this concept to chromosomes

and DNA. As shown in **Figure 14.4**, each gene is a sequence of nucleotides at a specific place, or locus, along a particular chromosome. The DNA at that locus, however, can vary slightly in its nucleotide sequence. This variation in information content can affect the function of the encoded protein and thus an inherited character of the organism. The purple-flower allele and the white-flower allele are two DNA sequence variations possible at the flower-color locus on a pea plant's chromosomes. The purple-flower allele sequence allows synthesis of purple pigment, and the white-flower allele sequence does not.

**Figure 14.4**

Alleles, alternative versions of a gene.

Homologous chromosomes with gene for flower color; arrows zoom in on D N A sequences for purple and white flower alleles. A functional enzyme is made from the purple flower allele only.

A pair of homologous chromosomes, one paternal and one maternal, with a region on each chromosome marked: an allele for purple flower color on the paternal chromosome and the allele for white flowers on the maternal chromosome. Arrows from each allele zoom in on the D N A molecules present on each chromosome at this location; the nucleotide sequence shown in the paternal (purple flower) allele is C T A A A T C G G T; the nucleotide sequence in the maternal (white flower) allele is A T A A A T C G G T. The D N A sequence in the paternal (purple flower) allele results, through a series of steps, in the production of an enzyme that helps synthesize purple pigment. The D N A sequence in the maternal (white flower) allele results in the absence of the enzyme. The interaction between these two alleles results in purple flower color because one purple flower allele results in sufficient pigment to produce purple flowers.

This diagram shows a pair of homologous chromosomes in an F1 hybrid pea plant, with the actual DNA sequence from the flower-color allele of each chromosome. The paternally inherited chromosome (blue) has an allele for purple flowers, which codes for a protein that indirectly controls synthesis of purple pigment. The maternally inherited chromosome (red) has an allele for white flowers, which results in no functional protein being made.

**Figure 14.4 Walkthrough**

Alleles, alternative versions of a gene

Watch an explanation of this figure in the Figure 14.4 Walkthrough: Alleles, alternative versions of a gene.

image

image

Second, *for each character, an organism inherits two versions (that is, two alleles) of a gene, one from each parent*. Remarkably, Mendel made this deduction without knowing about the role, or even the

existence, of chromosomes. Each somatic cell in a diploid organism has two sets of chromosomes, one set inherited from each parent (see Figure 13.4). Thus, a genetic locus is actually represented twice in a diploid cell, once on each homolog of a specific pair of chromosomes. The two alleles at a particular locus may be identical, as in the true-breeding plants of Mendel's P generation. Or the alleles may differ, as in the F1 hybrids (see Figure 14.4).

Third, *if the two alleles at a locus differ, then one, the* ***dominant allele****, determines the organism's appearance; the other, the* ***recessive allele****, has no noticeable effect on the organism's appearance*. Accordingly, Mendel's F1 plants had purple flowers because the allele for that trait is dominant and the allele for white flowers is recessive.

The fourth and final part of Mendel's model, the **law of segregation**, states that *the two alleles for a heritable character segregate (in other words, separate from each other) during gamete formation and end up in different gametes*. Thus, an egg or a sperm gets only one of the two alleles that are present in the diploid cells of the organism making the gamete. In terms of chromosomes, this segregation corresponds to the distribution of copies of the two members of a pair of homologous chromosomes to different gametes in meiosis (see Figure 13.7). Note that if an organism has identical alleles for a particular character, then that allele is present in all gametes. Because it is the only allele that can be passed on to offspring when the plant self-pollinates, the offspring always look the same as their parents in regard to that characteristic; this explains why these plants are true-breeding. But if different alleles are present, as in the F1 hybrids, then 50% of the gametes receive the dominant allele and 50% receive the recessive allele.

Does Mendel's segregation model account for the 3:1 ratio he observed in the F2 generation of his numerous crosses? For the flower-color character, the model predicts that the two different alleles present in an F1 individual will segregate into gametes such that half the gametes will have the purple-flower allele and half will have the white-flower allele. During self-pollination, gametes of each class unite randomly. An egg with a purple-flower allele has an equal chance of being fertilized by a sperm with a purple-flower allele or by a sperm with a white-flower allele. Since the same is true for an egg with a white-flower allele, there are four equally likely combinations of sperm and egg. **Figure 14.5** illustrates these combinations using a **Punnett square**, a handy diagrammatic device for predicting the allele composition of offspring from a cross between individuals of known genetic makeup. Notice that we use a capital letter to symbolize a dominant allele and a lowercase letter for a recessive allele. In our example, P is the purple-flower allele, and p is the white-flower allele; it is often useful as well to be able to refer to the gene itself as the P/p gene.

**Figure 14.5**

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Mendel's law of segregation.
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Mendel's law of segregation with 3 generations.

Each generation is as follows.

- P Generation. Each true-breeding plant of the parental generation has two identical alleles, denoted as either PP both uppercase (purple) or pp both lowercase (white). Purple and white flowers are crossed. Genetic makeup: PP both uppercase (purple) and pp both lowercase (white). Gametes (shown as circles) each contain only one allele for the flower-color gene. In this case, every gamete produced by a given parent has the same allele. Gametes: uppercase P for purple (both uppercase PP) flowers and lowercase p for white (pp both lowercase) flowers.
- F 1 Generation. All flowers appear purple. Genetic makeup: P p (uppercase P lowercase P). Gametes: one-half uppercase P and one-half lowercase p. Union of parental gametes produces F 1 hybrids having an uppercase P lowercase p combination. Because the purple-flower allele is dominant, all these hybrids have purple flowers. When the hybrid plants produce gametes, the two alleles segregate. Half of the gametes receive the uppercase P allele and the other half the lowercase p allele.
- F 2 Generation. Punnett square showing possible F 2 genotypes. Outside square are gametes from F 1 parents. Result is purple to white flowers in 3 to 1 ratio in the F 2 generation. This box, a Punnett square, shows all possible combinations of alleles in offspring that result from an F 1 by F 1 (uppercase P lowercase p by uppercase P lowercase p) cross. Each square represents an equally probable product of fertilization. For example, the bottom left box shows the genetic combination resulting from a lowercase p egg fertilized by a sperm. Random combination of the gametes results in the 3 to 1 ratio that Mendel observed in the F2 generation.

This diagram shows the genetic makeup of the generations in Figure 14.3. It illustrates Mendel's model for inheritance of the alleles of a single gene. Each plant has two alleles for the gene controlling flower color, one allele inherited from each of the plant's parents. To construct a Punnett square that predicts the F2 generation offspring, we list all the possible gametes from one parent (here, the F1 female) along the left side of the square and all the possible gametes from the other parent (here, the F1 male) along the top. The boxes represent the offspring resulting from all the possible unions of male and female gametes.

**Animation**

Mendel's Cross of One Character: Flower Color

image

image

**Animation**

Simplified Crosses of One Character in Humans

image

image

**Animation**

Cross of One Character in “MendAliens”

image

image

In the F2 offspring, what color will the flowers be? One-fourth of the plants have inherited two purple-flower alleles; clearly, these plants will have purple flowers. One-half of the F2 offspring have inherited one purple-flower allele and one white-flower allele; these plants will also have purple flowers, the dominant trait. Finally, one-fourth of the F2 plants have inherited two white-flower alleles and will express the recessive trait. Thus, Mendel’s model accounts for the 3:1 ratio of traits that he observed in the F2 generation.

## Useful Genetic Vocabulary

An organism that has a pair of identical alleles for a gene encoding a character is called a **homozygote** and is said to be **homozygous** for that gene. In the parental generation in Figure 14.5, the purple-flowered pea plant is homozygous for the dominant allele (PP), while the white plant is homozygous for the recessive allele (pp). Homozygous plants “breed true” because all of their gametes contain the same allele—either P or p in this example. If we cross dominant homozygotes with recessive homozygotes, every offspring will have two different alleles—Pp in the case of the F1 hybrids of our flower-color experiment (see Figure 14.5). An organism that has two different alleles for a gene is called a **heterozygote** and is said to be **heterozygous** for that gene. Unlike homozygotes, heterozygotes produce gametes with different alleles, so they are not true-breeding. For example, P- and p-containing gametes are both produced by our F1 hybrids. Self-pollination of the F1 hybrids thus produces both purple-flowered and white-flowered offspring.

Because of the different effects of dominant and recessive alleles, an organism’s traits do not always reveal its genetic composition. Therefore, we distinguish between an organism’s appearance or observable traits, called its **phenotype**, and its genetic makeup, its **genotype**. As shown in Figure 14.5 for the case of flower color in pea plants, PP and Pp plants have the same phenotype (purple flowers) but different genotypes. **Figure 14.6** reviews these terms. Note that the term *phenotype* refers not only to traits that relate directly to physical appearance but also to physiological traits. For example, one pea variety lacks the normal ability to self-pollinate, which is a phenotypic trait (called non-self-pollination).

**Figure 14.6**

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Phenotype versus genotype.
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Phenotypes are purple or white. Two different possible genotypes for purple flowers (PP both uppercase or uppercase P lowercase p) and one for white flowers (pp both lowercase).

Phenotype: In the F 2 offspring, there is a 3 to 1 ratio of purple to white flowers.

Genotype: in the F 2 offspring, there is a 1 to 2 to 1 ratio: 1 P P both uppercase (homozygous) to 2 uppercase P lowercase p (heterozygous) to 1 p p both lowercase (homozygous). The P P both uppercase and uppercase P lowercase p genotypes both result in purple flower phenotypes while the p p both lowercase genotype results in the white flower phenotype.

Grouping F2 offspring from a cross for flower color according to phenotype results in the typical 3:1 phenotypic ratio. In terms of genotype, however, there are actually two categories of purple-flowered plants, PP (homozygous) and PP (heterozygous), giving a 1:2:1 genotypic ratio.

### The Testcross

Given a purple-flowered pea plant, we cannot tell if it is homozygous PP or heterozygous Pp because both genotypes result in the same purple phenotype. To determine the genotype, we can cross this plant with a white-flowered plant pp, which will make only gametes with the recessive allele p. The allele in the gamete contributed by the purple-flowered plant of unknown genotype will therefore determine the appearance of the offspring **(Figure 14.7)**. If all the offspring of the cross have purple flowers, then the purple-flowered mystery plant must be homozygous for the dominant allele, because a PP×pp cross produces all Pp offspring. But if both the purple and the white phenotypes appear among the offspring, then the purple-flowered parent must be heterozygous. The offspring of a Pp×pp cross will be expected to have a 1:1 phenotypic ratio. Breeding an organism of unknown genotype with a recessive homozygote is called a **testcross** because it can reveal the genotype of that organism. The testcross was devised by Mendel and continues to be used by geneticists.

## Figure 14.7

## Research Method: The Testcross

### Application

An organism that shows a dominant trait in its phenotype, such as purple flowers in pea plants, can be either homozygous for the dominant allele or heterozygous. To determine the organism's genotype, geneticists can perform a testcross.

### Technique

In a testcross, the individual with the unknown genotype is crossed with a homozygous individual expressing the recessive trait (white flowers in this example), and Punnett squares are used to predict the possible outcomes.

A diagram of the experiment.

A purple flower is crossed with a white flower. The purple flower has dominant phenotype, unknown phenotype, uppercase P uppercase P or uppercase P lowercase p? White flower has recessive phenotype, known genotype, lower p lower p. Two Punnett squares are used to make predictions. First, if the purple flower parent is uppercase P lowercase P, the prediction is all purple flowers. If purple-flowered parent is uppercase P lowercase p, the prediction is half purple flowers and half white flowers.

### Results

Matching the results to either prediction identifies the unknown parental genotype (either PP or Pp in this example). In this testcross, we transferred pollen from a white-flowered plant to the carpels of a purple-flowered plant; the opposite (reciprocal) cross would have led to the same results.

All offspring purple or half offspring purple and half offspring white.

**Animation**

Testcross in “MendAliens”

image

image

## The Law of Independent Assortment

Mendel derived the law of segregation from experiments in which he followed only a *single* character, such as flower color. All the F1 progeny produced in his crosses of true-breeding parents were **monohybrids**, meaning that they were heterozygous for the one particular character being followed in the cross. We refer to a cross between such heterozygotes as a **monohybrid cross**.

Mendel worked out the second law of inheritance by following *two* characters at the same time, such as seed color and seed shape. Seeds (peas) may be either yellow or green. They also may be either round (smooth) or wrinkled. From single-character crosses, Mendel knew that the allele for yellow seeds is dominant (Y), and the allele for green seeds is recessive (y). For the seed-shape character, the allele for round is dominant (R), and the allele for wrinkled is recessive (r).

Imagine crossing two true-breeding pea varieties that differ in *both* of these characters—a cross between a plant with yellow round seeds (YYRR) and a plant with green wrinkled seeds (yyrr). The F1 plants will be **dihybrids**, individuals heterozygous for the two characters being followed in the cross (YyRr). But are these two characters transmitted from parents to offspring as a package? That is, will the Y and R alleles always stay together, generation after generation? Or are seed color and seed shape inherited independently? **Figure 14.8** shows how a **dihybrid cross**, a cross between F1 dihybrids, can determine which of these two hypotheses is correct.

# Figure 14.8

# Inquiry: Do the alleles for one character segregate into gametes dependently or independently of the alleles for a different character?

## Experiment

To follow the characters of seed color and seed shape through the F2 generation, Mendel crossed a true-breeding plant with yellow round seeds with a true-breeding plant with green wrinkled seeds, producing dihybrid F1 plants. Self-pollination of the F1 dihybrids produced the F2 generation. The two hypotheses (dependent and independent "assortment" of the two genes) predict different phenotypic ratios.

Two characteristics (seed shape and seed color) traced from parental generation to F 2 generation to test whether assortment is dependent or independent. Resulting phenotypic ratio of 9 to 3 to 3 to 1 supports independent assortment.

See table next tab

P generation: Round yellow seed (Y Y R R, all uppercase) crossed with wrinkled green seed (y y r r, all lowercase). Gametes from Y Y R R, all uppercase = 100% Y R, both uppercase and gametes from y y r r, all lowercase equals to 100% y r, both lowercase.

F1 generation equals to All round yellow seeds (all with genotypes uppercase Y lowercase y uppercase R lowercase r).

Hypothesis of dependent assortment predicts a phenotypic ratio of 3 round, yellow to 1 wrinkled, green. See table 14.8a.

Hypothesis of independent assortment predicts nine sixteenths round yellow seeds, three sixteenths round green seeds, three sixteenths wrinkled yellow seeds and one sixteenth wrinkled green seeds; or a phenotypic

ratio of 9 to 3 to 3 to 1.

| Gametes | one-fourth Y R, both uppercase | one-fourth uppercase Y lowercase r | uppercase R one-fourth lowercase y | one-fourth y r, both lowercase |
|---|---|---|---|---|
| one-fourth Y R, both uppercase | Y Y R R, all uppercase. Phenotype: Round, yellow | Y Y, both uppercase, uppercase R lowercase r. Phenotype: Round, yellow | Uppercase Y lowercase y, RR both uppercase. Phenotype: Round, yellow | Uppercase Y lowercase y Uppercase R lowercase r. Phenotype: Round, yellow |
| one-fourth uppercase Y lowercase r | Y Y, both uppercase, uppercase R lowercase r. Phenotype: Round, yellow | Y Y, both uppercase, rr both lowercase. Phenotype: wrinkled, yellow | Uppercase Y lowercase y Uppercase R lowercase r. Phenotype: Round, yellow | Uppercase Y lowercase y, rr, both lowercase. Phenotype: wrinkled, yellow |
| one-fourth lowercase y uppercase R | Uppercase Y lowercase y RR both uppercase. Phenotype: Round, yellow | Uppercase Y lowercase y Uppercase R lowercase r. Phenotype: Round, yellow | y y, both lowercase, RR both uppercase. Phenotype: Round, green. | y y, both lowercase, uppercase R lowercase r. Phenotype: Round, green. |
| one-fourth yr both lowercase | Uppercase Y lowercase y Uppercase R lowercase r. Phenotype: Round, yellow | Uppercase Y lowercase y, r r, both lowercase. Phenotype: wrinkled, yellow | y y, both lowercase, uppercase R lowercase r. Phenotype: Round, green. | y y r r, all lowercase. Phenotype: wrinkled, green. |

## Results

Results: 315 round yellow seeds, 108 round green seeds, 101 wrinkled yellow seeds and 32 wrinkled green seeds. Experimental phenotypic ratio = approximately 9 to 3 to 3 to 1.

| Gametes | one-half Y R, both uppercase | one-half y r, both lowercase |
|---|---|---|
| one half Y R, both uppercase | Y Y R R, all uppercase Phenotype: round, yellow | Uppercase Y lowercase y Uppercase R lowercase r. Phenotype: Round, Yellow |
| one half y r, both lowercase | Uppercase Y lowercase y Uppercase R lowercase r. Phenotype: round, yellow | y y r r all lowercase Phenotype: wrinkled, green |

## Conclusion

The results support the hypothesis of independent assortment, the only one that predicts two newly observed phenotypes: green round seeds and yellow wrinkled seeds (see the right-hand Punnett square). The alleles for each gene segregate independently of those of the other, and the two genes are said to assort independently.

**Data from** G. Mendel, Experiments in plant hybridization, *Proceedings of the Natural History Society of Brünn*4:3–47 (1866).

**WHAT IF?** *Suppose Mendel had transferred pollen from an*F1*plant to the carpel of a plant that was homozygous recessive for both genes. Set up the cross and draw Punnett squares that predict the offspring for both hypotheses. Would this cross have supported the hypothesis of independent assortment equally well?*

*For suggested answer, see*Appendix A.*

### Animation

Mendel's Cross of Two Characters: Seed Shape and Seed Color

image

image

### Animation

A Simplified Cross of Two Characters in Humans

image

image

### Animation

Crosses of Two Characters in “MendAliens”

image

The F1 plants, of genotype YyRr, exhibit both dominant phenotypes, yellow seeds with round shapes, no matter which hypothesis is correct. The key step in the experiment is to see what happens when F1 plants self-pollinate and produce F2 offspring. If the hybrids must transmit their alleles in the same combinations in which the alleles were inherited from the P generation, then the F1 hybrids will produce only two classes of gametes: YR and yr. As shown on the left side of Figure 14.8, this “dependent assortment” hypothesis predicts that the phenotypic ratio of the F2 generation will be 3:1, just as in a monohybrid cross:

Three-fourths yellow round, and one-fourth green wrinkled.

The alternative hypothesis is that the two pairs of alleles segregate independently of each other. In other words, genes are packaged into gametes in all possible allelic combinations, as long as each gamete has one allele for each gene (see Figure 13.11). In our example, an F1 plant will produce four classes of gametes in equal quantities: YR,Yr,yR, and yr. If sperm of the four classes fertilize eggs of the four classes, there will be 16(4×4) equally probable ways in which the alleles can combine in the F2 generation, as shown on the right side of Figure 14.8. These combinations result in four phenotypic categories with a ratio of 9:3:3:1 (9 yellow round to 3 green round to 3 yellow wrinkled to 1 green wrinkled):

Nine-sixteenths yellow ground to three-sixteenths green round to three-sixteenths yellow wrinkled to one-sixteenth green wrinkled.

When Mendel did the experiment and classified the F2 offspring, his results were close to the predicted 9:3:3:1 phenotypic ratio, supporting the hypothesis that the alleles for one gene—the gene for seed color, for example—segregate into gametes independently of the alleles of any other gene, such as the gene for seed shape.

Mendel tested his seven pea characters in various dihybrid combinations and always observed a 9:3:3:1 phenotypic ratio in the F2 generation. Is this consistent with the 3:1 phenotypic ratio seen for the monohybrid cross shown in Figure 14.5? To answer this question, count the number of yellow and green peas, ignoring shape, and calculate the ratio. The results of Mendel’s dihybrid experiments are the basis for what we now call the **law of independent assortment**, which states that *two or more genes assort independently—that is, each pair of alleles segregates independently of any other pair of alleles—during gamete formation.*

This law applies only to genes (allele pairs) located on different chromosomes (that is, on chromosomes that are not homologous) or, alternatively, to genes that are very far apart on the same chromosome. (This will be explained in Concept 15.3, along with the more complex inheritance patterns of genes located near each other, alleles of which tend to be inherited together.) All the pea characters Mendel chose for analysis were

controlled by genes on different chromosomes or were far apart on the same chromosome; this situation greatly simplified interpretation of his multicharacter pea crosses. All the examples we consider in the rest of this chapter involve genes located on different chromosomes.

**Animation**

Independent Assortment

## Concept Check 14.1

- **DRAW IT** Pea plants heterozygous for flower position and stem length (AaTt) are allowed to self-pollinate, and 400 of the resulting seeds are planted. Draw a Punnett square for this cross. How many offspring would be predicted to have terminal flowers and be dwarf? (See Table 14.1.)
- **WHAT IF?** List all gametes that could be made by a pea plant heterozygous for seed color, seed shape, and pod shape (YyRrli; see Table 14.1). How large a Punnett square would you need to draw to predict the offspring of a self-pollination of this "trihybrid"?
- **MAKE CONNECTIONS** In some pea plant crosses, the plants are self-pollinated. Is self-pollination considered asexual or sexual reproduction? Explain. (See Concept 13.1.)

*For suggested answers, see*Appendix A.**

As you learned in Figure 13.6b, meiosis in plants produces spores, not gametes. In flowering plants like the pea, each spore develops into a microscopic haploid gametophyte that contains only a few cells and is located on the parent plant. The gametophyte produces sperm, in pollen grains, and eggs, in the carpel. For simplicity, we will not include the gametophyte stage in our discussion of fertilization in plants.

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# Concept 14.2: Probability laws govern Mendelian inheritance

Mendel's laws of segregation and independent assortment reflect the same rules of probability that apply to tossing coins, rolling dice, and drawing cards from a deck. The probability scale ranges from 0 to 1. An event that is certain to occur has a probability of 1, while an event that is certain *not* to occur has a probability of 0. With a coin that has heads on both sides, the probability of tossing heads is 1, and the probability of tossing tails is 0. With a normal coin, the chance of tossing heads is 12, and the chance of tossing tails is 12. The probability of drawing the ace of spades from a 52-card deck is 152. The probabilities of all possible outcomes for an event must add up to 1. With a deck of cards, the chance of picking a card other than the ace of spades is 5152.

Tossing a coin illustrates an important lesson about probability. For every toss, the probability of heads is 12. The outcome of any particular toss is unaffected by what has happened on previous trials. We refer to phenomena such as coin tosses as independent events. Each toss of a coin, whether done sequentially with one coin or simultaneously with many, is independent of every other toss. And like two separate coin tosses, the alleles of one gene segregate into gametes independently of another gene's alleles (the law of independent assortment). We'll now look at two basic rules of probability that help us predict the outcome of the fusion of such gametes in simple monohybrid crosses and more complicated crosses as well.

## The Multiplication and Addition Rules Applied to Monohybrid Crosses

How do we determine the probability that two or more independent events will occur together in some specific combination? For example, what is the chance that two coins tossed simultaneously will both land heads up? The **multiplication rule** states that to determine the probability of one event *and* the other occurring, we multiply the probability of one event (one coin coming up heads) by the probability of the other event (the other coin coming up heads). By the multiplication rule, then, the probability that both coins will land heads up is 12×12=14.

We can apply the same reasoning to an F1 monohybrid cross. With seed shape in pea plants as the heritable character, the genotype of F1 plants is Rr. Segregation in a heterozygous plant is like flipping a coin in terms of calculating the probability of each outcome: Each egg produced has a 12 chance of carrying the dominant allele (R) and a 12 chance of carrying the recessive allele (r). The same odds apply to each sperm cell produced. For a particular F2 plant to have wrinkled seeds, the recessive trait, both the egg and the sperm that come together must carry the r allele. The probability that an r allele will be present in the egg *and* in the sperm at fertilization is found by multiplying 12 (the probability that the egg will have an r)×12 (the probability that the sperm will have an r). Thus, the multiplication rule tells us that the probability of an F2 plant having wrinkled seeds (rr) is 14**(**Figure 14.9**)**. Likewise, the probability of an F2 plant carrying both dominant alleles for seed shape (RR) is 14.

**Figure 14.9**

Segregation of alleles and fertilization as chance events.

Probabilities of formation and fertilization of gametes (uppercase R or lowercase r) from two heterozygous parents compared to toss of a coin (heads or tails) in a Punnett square.

| Gametes | one-half uppercase R (heads) | one-half lowercase r (tails) |
|---|---|---|

| | | |
|---|---|---|
| one-half uppercase R (heads) | RR both uppercase, both coins are heads | Uppercase R lowercase |
| | Probability = one-fourth | One heads, one tails |
| | | Probability = one-fourth |
| one-half lowercase r (tails) | Uppercase R lowercase r | rr both lowercase, both coins are tails |
| | One heads, one tails | Probability = one-fourth |
| | Probability = one-fourth | |

Alleles from upper case R lowercase r parent segregate into eggs, while alleles from another uppercase R lowercase r parent segregate into sperm. Half of the gametes from each parent will be uppercase R and half will be lowercase r, just as there is a one-half chance of flipping heads and a one-half chance of flipping tails when flipping a coin. The gametes (eggs and sperm) are superimposed on coins (heads or tails) in a Punnett square with two rows and two columns. The eggs (left side) are one-half uppercase R (heads) and one-half lowercase r (tails). The sperm (top) are one-half uppercase R (heads) and one-half lowercase r (tails). A Punnett square is represented by the following table.

When a heterozygote (Rr) forms gametes, whether a particular gamete ends up with an R or an r is like the toss of a coin. We can determine the probability for any genotype among the offspring of two heterozygotes by multiplying together the individual probabilities of an egg and sperm having a particular allele (R or r in this example).

To figure out the probability that an F2 plant from a monohybrid cross will be heterozygous rather than homozygous, we need to invoke a second rule. Notice in Figure 14.9 that the dominant allele can come from the egg and the recessive allele from the sperm, or vice versa. That is, F1 gametes can combine to produce Rr offspring in two *mutually exclusive* ways: For any particular heterozygous F2 plant, the dominant allele can come from the egg *or* the sperm, but not from both. According to the **addition rule**, the probability that any one of two or more mutually exclusive events (one event *or* the other) will occur is calculated by adding their individual probabilities. As we have just seen, the multiplication rule gives us the individual probabilities that we will now add together. The probability for one possible way of obtaining an F2 heterozygote—the dominant allele from the egg and the recessive allele from the sperm—is 14. The probability for the other possible way—the recessive allele from the egg and the dominant allele from the sperm—is also 14 (see Figure 14.9). Using the rule of addition, then, we can calculate the probability of an F2 heterozygote as 14+14=12. (Note that use of the word *and* in the statement is a clue that you should use the multiplication rule, while *or* suggests use of the addition rule.)

## Solving Complex Genetics Problems with the Rules of Probability

We can also apply the rules of probability to predict the outcome of crosses involving multiple characters. Recall that each allelic pair (gene) segregates independently during gamete formation (the law of independent assortment). Thus, a dihybrid or other multicharacter cross is equivalent to two or more independent monohybrid crosses occurring simultaneously. By applying what we have learned about monohybrid crosses, we can determine the probability of specific genotypes occurring in the F2 generation without having to construct unwieldy Punnett squares.

Consider the dihybrid cross between YyRr heterozygotes shown in Figure 14.8. We will focus first on the seed-color character. For a monohybrid cross of Yy plants, we can use a simple Punnett square to determine that the probabilities of the offspring genotypes are 14 for YY, 12 for Yy, and 14 for yy. We can draw a second Punnett square to determine that the same probabilities apply to the offspring genotypes for seed shape: 14 RR,12Rr, and 14rr. Knowing these probabilities, we can simply use the multiplication rule to determine the probability of each of the genotypes in the F2 generation. To give two examples, the calculations for finding the probabilities of two of the possible F2 genotypes (YYRR and YyRR) are shown below:

Probability of upper Y Y R R = one fourth, probability of Y Y, times one fourth, R R, = one sixteenth. Probability of upper Y lower y R R = one half, upper Y lower y, times one fourth, R R, = one eighth.

The YYRR genotype corresponds to the upper left box in the larger Punnett square in Figure 14.8 (one box=116). Looking closely at the larger Punnett square in Figure 14.8, you will see that 2 of the 16 boxes (18) correspond to the YyRR genotype.

Now let's see how we can combine the multiplication and addition rules to solve even more complex problems in Mendelian genetics. Imagine a cross of two pea varieties in which we track the inheritance of three characters. Let's cross a trihybrid with purple flowers and yellow round seeds (heterozygous for all three genes) with a plant with purple flowers and green wrinkled seeds (heterozygous for flower color but homozygous recessive for the other two characters). Using Mendelian symbols, our cross is PpYyRr×Ppyyrr. What fraction of offspring from this cross are predicted to exhibit the recessive phenotypes for *at least two* of the three characters?

To answer this question, we can start by listing all genotypes we could get that fulfill this condition: ppyyRr, ppYyrr, Ppyyrr*,*PPyyrr, and ppyyrr. (Because the condition is *at least two* recessive traits, it includes the last genotype, which shows all three recessive traits.) Next, we calculate the probability for each of these genotypes resulting from our PpYyRr×Ppyyrr cross by multiplying together the individual probabilities for the allele pairs, just as we did in our dihybrid example. Note that in a cross involving heterozygous and homozygous allele pairs (for example, Yy×yy), the probability of heterozygous (Yy) offspring is 12 and the

probability of homozygous (in this case, yy) offspring is 12. Finally, we use the addition rule to add the probabilities for all the different genotypes that fulfill the condition of at least two recessive traits resulting from our PpYyRr×Ppyyrr cross, as shown below:

Five probability calculations.

The calculations are as follows.

probability lower p p y y upper R lower r = one fourth, probability of lower p p, times one half, lower y y, times one half uppercase R lower r equals to one-sixteenth.

probability lower p p upper Y lowercase y r r equals to one fourth, lower p p, times one half, uppercase Y lowercase y, times one half, lowercase r r equals to one-sixteenth.

probability uppercase P lowercase p y y r r equals to one half, uppercase P lowercase p, times one half lowercase y y, times one half lowercase r r equals to two sixteenths.

probability uppercase P P lowercase y y r r equals to one-fourth, upper P P, times one half, lowercase y y, times one half, lowercase r r, equals to one-sixteenth.

probability lowercase p p y y r r equals to one-fourth, lowercase p p, times one-half, lowercase y y, times one half, lowercase r r equals to one-sixteenth.

In time, you'll be able to solve genetics problems faster by using the rules of probability than by filling in Punnett squares.

We cannot predict with certainty the exact numbers of progeny of different genotypes resulting from a genetic cross. But the rules of probability give us the *likelihood* of various outcomes. Usually, the larger the sample size, the closer the results will conform to our predictions. Mendel understood this statistical feature of inheritance and had a keen sense of the rules of chance. It was for this reason that he set up his experiments so as to generate, and then count, large numbers of offspring from his crosses.

## Concept Check 14.2

- For any gene with a dominant allele A and recessive allele a, what proportions of the offspring from an AA×Aa cross are expected to be homozygous dominant, homozygous recessive, and heterozygous?
- Two organisms, with genotypes BbDD and BBDd, are mated. Assuming independent assortment of the B/b and D/d genes, write the genotypes of all possible offspring from this cross and use the rules of probability to calculate the chance of each genotype occurring.
- **WHAT IF?** Three characters (flower color, seed color, and pod shape) are considered in a cross between two pea plants: PpYyIi×ppYyIi. Using the rules of probability, determine the fraction of

offspring predicted to be homozygous recessive for at least two of the three characters.

*For suggested answers, see*Appendix A.**

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# Concept 14.3: Inheritance patterns are often more complex than predicted by simple Mendelian genetics

In the 1900s, geneticists extended Mendelian principles not only to diverse organisms, but also to patterns of inheritance more complex than those described by Mendel. For the work that led to his two laws of inheritance, Mendel chose pea plant characters that turn out to have a relatively simple genetic basis: Each character is determined by one gene, for which there are only two alleles, one completely dominant and the other completely recessive. (There is one exception: Mendel's pod shape character is actually determined by two genes.) Not all heritable characters are determined so simply, and the relationship between genotype and phenotype is rarely so straightforward. Mendel himself realized that he could not explain the more complicated patterns he observed in crosses involving other pea characters or other plant species. This does not diminish the utility of Mendelian genetics, however, because the basic principles of segregation and independent assortment apply even to more complex patterns of inheritance. In this section, we will extend Mendelian genetics to hereditary patterns that were not reported by Mendel.

## Extending Mendelian Genetics for a Single Gene

The inheritance of characters determined by a single gene deviates from simple Mendelian patterns when alleles are not completely dominant or recessive, when a particular gene has more than two alleles, or when a single gene produces multiple phenotypes. We will describe examples of each of these situations in this section.

### Degrees of Dominance

Alleles can show different degrees of dominance and recessiveness in relation to each other. In Mendel's classic pea crosses, the F1 offspring always looked like one of the two parental varieties because one allele in a pair showed **complete dominance** over the other. In such situations, the phenotypes of the heterozygote and the dominant homozygote are indistinguishable (see Figure 14.6).

For some genes, however, neither allele is completely dominant, and the F1 hybrids have a phenotype somewhere between those of the two parental varieties. This phenomenon, called **incomplete**

**dominance**, is seen when red snapdragons are crossed with white snapdragons: All the F1 hybrids have pink flowers **(Figure 14.10)**. This third, intermediate phenotype results from flowers of the heterozygotes having less red pigment than the red homozygotes. (This is unlike the case of Mendel's pea plants, where the Pp heterozygotes make enough pigment for the flowers to be purple, indistinguishable from those of PP plants.)

**Figure 14.10**

```
Incomplete dominance in snapdragon color.
```

P Generation = Red flower crossed with white flower. F 1 Generation = all pink flowers. F 2 Generation = 1 Red to 2 pink to 1 white flowers.

| Gametes | one-half C superscript R | one-half C superscript W |
|---|---|---|
| one-half C superscript R | C superscript R, C superscript R. Phenotype: Red flower | C superscript R, C superscript W. Phenotype: Pink flower |
| one-half C superscript W | C superscript R, C superscript W. Phenotype: Pink flower | C superscript W, C superscript W. Phenotype: Pink flower |

P Generation: Red Flower (C superscript R, C superscript R) x White Flower (C superscript W, C superscript W). Arrows indicate red flower gamete (C superscript R) crossed with white flower gamete (C superscript W), flowing to F 1 Generation.

F 1 Generation: All pink flowers (C superscript W, C superscript W). Arrows indicate gametes are one-half C superscript R and one-half C superscript W. F 2 Generation: Punnett square with eggs on the left side and sperm on top. A Punnett square is represented by the following table.

| Gametes | one-half C superscript R | one-half C superscript W |
|---|---|---|
| one-half C superscript R | C superscript R, C superscript R. Phenotype: Red flower | C superscript R, C superscript W. Phenotype: Pink flower |
| one-half C superscript W | C superscript R, C superscript W. Phenotype: Pink flower | C superscript W, C superscript W. Phenotype: Pink flower |

When red snapdragons are crossed with white ones, the F1 hybrids have pink flowers. Segregation of alleles into gametes of the F1 plants results in an F2 generation with a 1:2:1 ratio for both genotype and phenotype.

Neither allele is dominant, so rather than using upper- and lowercase letters, we use the letter *C* with a superscript to indicate an allele for flower color: CR for red and CW for white.

*Suppose a classmate argues that this figure supports the blending hypothesis for inheritance. What might your classmate say, and how would you respond?*

*For suggested answer, see*Appendix A.**

**Animation**

Incomplete Dominance in “MendAliens”

image

Play

image

Play

At first glance, incomplete dominance of either allele seems to provide evidence for the blending hypothesis of inheritance, which would predict that the red or white trait could never reappear among offspring of the pink hybrids. In fact, interbreeding F1 hybrids produces F2 offspring with a phenotypic ratio of one red to two pink to one white. (Because heterozygotes have a separate phenotype from homozygotes, the genotypic and phenotypic ratios for the F2 generation are the same, 1:2:1.) The segregation of the red-flower and white-flower alleles in the gametes produced by the pink-flowered plants confirms that the alleles for flower color are heritable factors that maintain their identity in the hybrids; that is, that the factors are discrete rather than “blendable.”

Another variation on dominance relationships between alleles is called **codominance**; in this variation, the two alleles each affect the phenotype in separate, distinguishable ways. For example, the human MN blood group is determined by codominant alleles for two specific molecules located on the surface of red blood cells, the M and N molecules. A single gene (L), for which two allelic variations are possible (LM or LN), determines the phenotype of this blood group. Individuals homozygous for the LM allele LMLM have red blood cells with only M molecules; individuals homozygous for the LN allele LNLN have red blood cells with only N molecules. But *both* M and N molecules are present on the red blood cells of individuals heterozygous for the M and N alleles LMLN. Note that the MN phenotype is *not* intermediate between the M and N phenotypes, which distinguishes codominance from incomplete dominance. Rather, *both* M and N phenotypes are exhibited by heterozygotes, since both molecules are present.

## The Relationship Between Dominance and Phenotype

We've now seen that the relative effects of two alleles range from complete dominance of one allele to incomplete dominance of either allele to codominance of both alleles. It is important to understand that an allele is called *dominant* because it is seen in the phenotype, not because it somehow subdues a recessive allele. Alleles are simply variations in a gene's nucleotide sequence (see Figure 14.4). When a dominant allele coexists with a recessive allele in a heterozygote, they do not actually interact at all. It is in the pathway from genotype to phenotype that dominance and recessiveness come into play.

To illustrate the relationship between dominance and phenotype, we can use one of the characters Mendel studied—round versus wrinkled pea seed shape. The dominant allele (round) codes for an enzyme that helps convert an unbranched form of starch to a branched form in the seed. The recessive allele (wrinkled) codes for a defective form of this enzyme, leading to an accumulation of unbranched starch, which causes excess water to enter the seed by osmosis. Later, when the seed dries, it wrinkles. If a dominant allele is present, no excess water enters the seed and it does not wrinkle when it dries. One dominant allele results in enough of the enzyme to synthesize adequate amounts of branched starch, which means that dominant homozygotes and heterozygotes have the same phenotype: round seeds.

A closer look at the relationship between dominance and phenotype reveals an intriguing fact: For any character, the observed dominant/recessive relationship of alleles depends on the level at which we examine phenotype. **Tay-Sachs disease**, an inherited disorder in humans, is an example. The brain cells of a child with Tay-Sachs disease cannot metabolize certain lipids because a crucial enzyme does not work properly. As these lipids accumulate in brain cells, the child begins to experience seizures, blindness, and degeneration of motor and mental performance and dies within a few years.

Only children who inherit two copies of the Tay-Sachs allele (homozygotes) have the disease. Thus, at the *organismal* level, the Tay-Sachs allele qualifies as recessive. However, the activity level of the lipid-metabolizing enzyme in heterozygotes is intermediate between the activity level in individuals homozygous for the normal allele and the activity level in individuals with Tay-Sachs disease. (The term *normal* is used in the genetic sense to refer to the allele coding for the enzyme that functions properly.) The intermediate phenotype observed at the *biochemical* level is characteristic of incomplete dominance of either allele. Fortunately, the heterozygote condition does not lead to disease symptoms, apparently because half the normal enzyme activity is sufficient to prevent lipid accumulation in the brain. Extending our analysis to yet another level, we find that heterozygous individuals produce equal numbers of normal and dysfunctional enzyme molecules. Thus, at the *molecular* level, the normal allele and the Tay-Sachs allele are codominant. As you can see, whether alleles appear to be completely dominant, incompletely dominant, or codominant depends on the level at which the phenotype is analyzed.

## Frequency of Dominant Alleles

While you might assume that the dominant allele for a particular character would be more common than the recessive allele, this is not always the case. For an example of a rare dominant allele, about one baby out of

400 in the United States is born with extra fingers or toes, a condition known as polydactyly. Some cases are caused by the presence of a dominant allele. The low frequency of these types of polydactyly indicates that the recessive allele, which results in five digits per appendage, is far more prevalent than the dominant allele in the population.

## Multiple Alleles

Only two alleles exist for the pea characters that Mendel studied, but most genes have more than two alleles. The ABO blood groups in humans, for instance, are determined by the two alleles a person has of the blood group gene; the three possible alleles are IA, IB, and i. A person's blood group may be one of four types: A, B, AB, or O. These letters refer to two carbohydrates—A and B—that may be found attached to specific cell-surface molecules on red blood cells. An individual's blood cells may have carbohydrate A (type A blood), carbohydrate B (type B), both (type AB), or neither (type O), as shown in **Figure 14.11**, along with the relevant genotypes. Matching compatible blood groups is critical for safe blood transfusions (see Concept 43.3).

**Figure 14.11**

Multiple alleles for the ABO blood groups.

Multiple alleles for the A B O blood groups.

*The four blood groups result from different combinations of three alleles.*

| Allele | upper I superscript A | upper I superscript B | lower i |
|---|---|---|---|
| Carbohydrate | A, triangle | B, circle | none |

Figure a is a table for the 3 alleles for the A B O blood groups and their carbohydrates. Each allele codes for an enzyme that may add a specific carbohydrate (designated by the superscript on the allele and shown as a triangle or circle) to red blood cells.

The table is as follows:

Figure b is of blood group genotypes and phenotypes. There are six possible genotypes, resulting in four different phenotypes. The table is as follows:

| Genotype | upper I superscript A, upper I superscript A, or | upper I superscript B, upper I superscript B, or | upper I superscript A, upper I superscript B | lower i, lower i |
|---|---|---|---|---|

| | upper I superscript A, lower i | upper I superscript B, lower i | | |
|---|---|---|---|---|
| Red blood cell with surface carbohydrates | red blood cell with triangles on its surface | red blood cell with circles on its surface | red blood cell with triangles and circles | red blood cell with nothing on its surface |
| Phenotype (blood group) | A | B | A B | o |

**VISUAL SKILLS** *Based on the surface carbohydrate phenotype in part (b), what are the dominance relationships among the alleles?*

*For suggested answer, see*Appendix A.**

### Pleiotropy

So far, we have treated Mendelian inheritance as though each gene affects only one phenotypic character. Most genes, however, have multiple phenotypic effects, a property called **pleiotropy** (from the Greek *pleion*, more). In humans, for example, pleiotropic alleles are responsible for the multiple symptoms associated with certain hereditary diseases, such as cystic fibrosis and sickle-cell disease, discussed later in this chapter. In the garden pea, the gene that determines flower color also affects the color of the coating on the outer surface of the seed, which can be gray or white. Given the intricate molecular and cellular interactions responsible for an organism's development and physiology, it isn't surprising that a single gene can affect a number of characters.

## Extending Mendelian Genetics for Two or More Genes

Dominance relationships, multiple alleles, and pleiotropy all have to do with the effects of the alleles of a single gene. We now consider two situations in which two or more genes are involved in determining a particular phenotype. In the first case, called epistasis, one gene affects the phenotype of another because the two gene products interact; in the second case, called polygenic inheritance, multiple genes independently affect a single trait.

### Epistasis

In **epistasis** (from the Greek for “standing upon”), the phenotypic expression of a gene at one locus alters that of a gene at a second locus. An example will help clarify this concept. In Labrador retrievers (commonly called “Labs”), black coat color is dominant to brown. Let’s designate B and b as the two alleles for this character. For a Lab to have brown fur, its genotype must be bb; these dogs are called chocolate Labs. But there is more to the story. A second gene determines whether or not pigment will be deposited in the hair. The dominant allele, symbolized by E, results in the deposition of either black or brown pigment, depending on the genotype at the first locus. But if the Lab is homozygous recessive for the second locus ee, then the coat is yellow, regardless of the genotype at the black/brown locus (yellow Labs). In this case, the gene for pigment deposition E/e is said to be *epistatic to* the gene that codes for black or brown pigment B/b.

What happens if we mate black Labs that are heterozygous for both genes BbEe? Although the two genes affect the same phenotypic character (coat color), they follow the law of independent assortment. Thus, our breeding experiment represents an F1 dihybrid cross, like those that produced a 9:3:3:1 ratio in Mendel’s experiments. We can use a Punnett square to represent the genotypes of the F2 offspring **(Figure 14.12)**. As a result of epistasis, the phenotypic ratio among the F2 offspring is 9 black to 3 chocolate to 4 yellow Labs. Other types of epistatic interactions produce different ratios, but all are modified versions of 9:3:3:1.

**Figure 14.12**

```
An example of epistasis.
```

Black labs (uppercase B lowercase b uppercase E lowercase e) mated; results shown in a Punnett square with 16 boxes. 9 black, 3 chocolate and 4 golden labs result.

Figure b is of blood group genotypes and phenotypes. There are six possible genotypes, resulting in four different phenotypes. The table is as follows:

| Genotype | upper I superscript A, upper I superscript A, or upper I superscript A, lower i | upper I superscript B, upper I superscript B, or upper I superscript B, lower i | upper I superscript A, upper I superscript B | lower i, lower i |
|---|---|---|---|---|
| Red blood cell with surface carbohydrates | red blood cell with triangles on its surface | red blood cell with circles on its surface | red blood cell with triangles and circles | red blood cell with nothing |

| | | | | |
|---|---|---|---|---|
| | | | | on its surface |
| Phenotype (blood group) | A | B | A B | O |

This Punnett square illustrates the genotypes and phenotypes predicted for offspring of matings between two black Labrador retrievers of genotype BbEe. The E/e gene, which is epistatic to the B/b gene coding for hair pigment, controls whether or not pigment of any color will be deposited in the hair.

**VISUAL SKILLS** *Compare the four squares in the lower right of this Punnett square with those in*Figure 14.8*.* Explain the genetic basis for the difference between the ratio*(9:3:4)*of phenotypes seen in this cross and the*9:3:3:1*ratio seen in*Figure 14.8*.*

*For suggested answer, see Appendix A.*

## Polygenic Inheritance

Mendel studied characters that could be classified on an either-or basis, such as purple versus white flower color. But many characters, such as human skin color and height, are not one of two discrete characters, but instead vary in the population in gradations along a continuum. These are called **quantitative characters**. Quantitative variation usually indicates **polygenic inheritance**, an additive effect of two or more genes on a single phenotypic character. (In a way, this is the converse of pleiotropy, where a single gene affects several phenotypic characters.) Height is a good example of polygenic inheritance: In 2014, a genomic study of over 250,000 people found almost 700 genetic variations in over 180 genes that affect height. Many variations were in or near genes involved in biochemical pathways affecting growth of the skeleton, but others were associated with genes not obviously related to growth. Another study in 2018, of 300,000 individuals, identified 124 genes that affect hair color. Although simplifying can sometimes be valuable in understanding genetic principles, as we'll see shortly, the classic idea of a single gene determining eye color or hair color or other such traits is an oversimplification. (Earlobe attachment, an example used until a decade ago in this textbook, was shown in 2017 to be affected by nearly 50 genes!)

Skin pigmentation in humans is also controlled by many separately inherited genes—378 at latest count, many of which are involved in the production of melanin skin pigments. Here, we'll simplify the story in order to understand the concept of polygenic inheritance. Let's consider three genes, with a dark-skin allele for each gene (A, B, or C) contributing one "unit" of darkness (also a simplification) to the phenotype and being incompletely dominant to the other, light-skin allele (a, b, or c). In our model, an AABBCC person would be very dark, whereas an aabbcc individual would be very light. An AaBbCc person would have skin of an intermediate shade. Because the alleles have a cumulative effect, the genotypes AaBbCc and AABbcc would make the same genetic contribution (three units) to skin darkness. There are seven skin color

phenotypes that could result from a mating between AaBbCc heterozygotes, as shown in **Figure 14.13**. In a large number of such matings, the majority of offspring would be expected to have intermediate phenotypes (skin color in the middle range). You can graph the predictions from the Punnett square in the **Scientific Skills Exercise**. Environmental factors, such as exposure to the sun, also affect the skin color phenotype.

**Figure 14.13**

```
A simplified model for polygenic inheritance of skin color.
```

Possible combinations of alleles (genes A, B, C) for skin color between parents heterozygous for all three genes. Skin colors range across seven phenotypes from very light to very dark.

Parental genotypes for two black labs: both uppercase B lowercase b uppercase E lowercase e.

| Gametes | one-fourth uppercase B and E | one-fourth lowercase b uppercase E | one-fourth Uppercase B lowercase e | one-fourth lowercase b and e |
|---|---|---|---|---|
| one-fourth uppercase B and E | Uppercase B, B, E, E (black lab) | Uppercase B, lowercase b, Uppercase E, E (black lab) | Uppercase B, B, E, lowercase e (black lab) | Uppercase B, lowercase b, Uppercase E, lowercase e (black lab) |
| one-fourth lowercase b uppercase E | Uppercase B, lowercase b, upper case E, E (black lab) | lowercase b, b, Uppercase E, E (brown lab) | Uppercase B, lowercase b, uppercase E, lowercase e (black lab) | lowercase b, b, Uppercase E, lowercase e (brown lab) |
| one-fourth Uppercase B lowercase e | Uppercase B, B, E, lowercase e (black lab) | Uppercase B, lowercase b, Uppercase E, lowercase e (black lab) | Uppercase B, B, lowercase e, e (golden lab) | Uppercase B, lowercase b, lowercase e, e (golden lab) |
| one-fourth lowercase b and e | Uppercase B, lower case b, uppercase E, | lowercase b, b, Uppercase E, | Uppercase B, lowercase b, e, e (golden lab) | lowercase b, b, e, e (golden lab) |

| | | | | |
|---|---|---|---|---|
| | lowercase e (black lab) | lowercase e (brown lab) | | |

| Gametes | one eighth a b c all lowercase | one eighth Uppercase A, lowercase b c | one eighth Lowercase a uppercase B lowercase c | one eighth Lowercase a lowercase b uppercase C | one eighth Uppercase A uppercase B lowercase c | one e Uppe A lower b uppe C |
|---|---|---|---|---|---|---|
| one eighth a b c all lowercase | a a b b c c all lowercase | Uppercase A, a b b cc all lowercase | Lowercase a a, uppercase B lowercase b, cc both lowercase | a a b b all lowercase, uppercase C lowercase c | Uppercase A lowercase a uppercase B, b c c all lowercase | Upper A, a b lowerc upper lowerc |
| one eighth uppercase A lowercase b c | Uppercase A, a b b c c all lowercase | AA both uppercase, b b c c all lowercase | Uppercase A lowercase a uppercase B, bcc all lowercase | Uppercase A, a b b all lowercase, uppercase C lowercase c | AAB all uppercase, bcc all lowercase | AA bo upper b b bo lowerc upper lowerc |
| one eighth lowercase a uppercase B lowercase c | aa both uppercase, uppercase B, bcc all lowercase | Uppercase A lowercase a uppercase B, bcc all lowercase | aa both lowercase, BB both uppercase, cc both lowercase | aa both lowercase, uppercase B lowercase b uppercase C lowercase c | Uppercase A lowercase a, BB both uppercase, cc both lowercase | Upper A lowe a uppe B lowe b uppe C lowe c |
| one eighth ab both lowercase, | a a b b all lowercase, uppercase | Uppercase A, a b b all lowercase, | aa both lowercase, uppercase B lowercase b | a a b b all lowercase, C C both uppercase | Uppercase A lowercase a uppercase B lowercase | Upper A, a b lowerc |

| | | | | | | |
|---|---|---|---|---|---|---|
| uppercase C | C lowercase c | uppercase C lowercase c | uppercase C lowercase c | | b uppercase C lowercase c | C C bo upper |
| one eighth A B both uppercase, lowercase c | Uppercase A lowercase a uppercase B, b c c all lowercase | A A B all uppercase, bcc all lowercase | Uppercase A lowercase a, B B both uppercase, c c both lowercase | Uppercase A lowercase a uppercase B lowercase b uppercase C lowercase c | A A B B all uppercase, cc both lowercase | A A B upper lowerc upper lowerc |
| one eighth uppercase A lowercase b uppercase C | Uppercase A, a b b all lowercase, uppercase C lowercase c | AA both uppercase, bb both lowercase, uppercase C lowercase c | Uppercase A lowercase a uppercase B lowercase b uppercase C lowercase c | Uppercase A, a b b all lowercase, C C both uppercase | A A B all uppercase, lowercase b uppercase C lowercase c | AA bo upper bb bot lowerc C C bo upper |
| one eighth lowercase a uppercase BC | aa both lowercase, uppercase B lowercase b uppercase C lowercase c | Uppercase A lowercase a uppercase B lowercase b uppercase C lowercase c | aa both lowercase, uppercase B lowercase b uppercase C lowercase c | aa both lowercase uppercase B lowercase b, CC both uppercase | Uppercase A lowercase a, BBC all uppercase, lowercase c | Upper A lowe a uppe B lowe b, CC l upper |
| one eighth A B C all uppercase | Uppercase A lowercase a uppercase B lowercase b uppercase C lowercase c | A A B all uppercase, lowercase b uppercase C lowercase c | Uppercase A lowercase a, B B C all uppercase, lowercase c | Uppercase A lowercase a uppercase B lowercase b, C C both uppercase | A A B B C all uppercase, lowercase c | A A B upper lowerc C C bo upper |

There are 7 skin phenotypes shown, with increasing darkness in skin color related to the number of dark-skin alleles present (0 to 6).

one sixty-fourth phenotypes are very light with no dark-skin alleles.

six sixty fourths phenotypes are light with one dark-skin allele.

fifteen sixty-fourths phenotypes are slightly darker, with 2 dark-skin alleles.

twenty sixty fourths phenotypes are slightly darker, with 3 dark-skin alleles.

fifteen sixty thirds phenotypes are brown with 4 dark-skin alleles.

six sixty fourths phenotypes are darker brown with 5 dark skin alleles.

one sixty fourth phenotypes are dark brown with 6 dark-skin alleles.

In this model, three separately inherited genes affect skin color. (The reported number is actually 378 genes.) The heterozygous individuals (AaBbCc) represented by the two rectangles at the top of this figure each carry three dark-skin alleles (black circles, which represent A, B, or C) and three light-skin alleles (white circles, which represent a, b, or c). The Punnett square shows all possible genetic combinations in gametes and offspring of hypothetical matings between these two. The results are summarized by phenotypic frequencies (fractions) under the Punnett square. (The phenotypic ratio is 1:6:15:20:15:6:1.)

**HHMI BioInteractive Video**

The Biology of Skin Color

image

Play

image

Play

For more resources from HHMI BioInteractive, go to [www.biointeractive.org](http://www.biointeractive.org).

## Scientific Skills Exercise

### *Making a Histogram and Analyzing a Distribution Pattern*

## AP Science Practices 2, 4

**What Is the Distribution of Phenotypes Among Offspring of Two Parents Who Are Both Heterozygous for Three Additive Genes?** Human skin color is a polygenic trait that is determined by the additive effects of many different genes. In this exercise, you will work with a simplified model of skin color genetics where only three genes are assumed to affect the darkness of skin color and where each gene has two alleles—dark or light (see Figure 14.13). In this model, each dark allele contributes equally to the darkness of skin color, and each pair of alleles segregates independently of any other pair. Using a type of graph called a histogram, you will determine the distribution of phenotypes of offspring with different numbers of dark-skin alleles. (For additional information about graphs, see the Scientific Skills Review in Appendix D.)

A group of people with different colored hair, eyes, and skin.

**How This Model Is Analyzed** To predict the phenotypes of the offspring of parents heterozygous for the three genes in our simplified model, we can use the Punnett square in Figure 14.13. The heterozygous individuals (AaBbCc) represented by the two rectangles at the top of that figure each carry three dark-skin alleles (black circles, which represent A, B, or C) and three light-skin alleles (white circles, which represent a, b, or c). The Punnett square shows all the possible genetic combinations in gametes and in offspring of a large number of hypothetical matings between these heterozygotes.

**Predictions from the Punnett Square** If we assume that each square in the Punnett square represents one offspring of the heterozygous AaBbCc parents, then the squares below show the possible skin color phenotypes and their predicted frequencies. Below the squares is the number of dark-skin alleles for each phenotype.

A table lists the phenotypes and the number of dark-skin alleles.

The table is as follows:

| 1 over 64 | 6 over 64 | 15 over 64 | 20 over 64 | 15 over 64 | 6 over 64 | 1 over 64 |
|---|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 |

Parents both normal phenotypes, and both with genotypes uppercase A lowercase a.

Probabilities for phenotypes in their children are one-fourth normal, one-half normal carriers, and one-fourth albino.

| Gametes | Uppercase A | Lowercase a |
|---|---|---|

| | | |
|---|---|---|
| Uppercase A | A A both uppercase Phenotype: normal | Uppercase A lower case a Phenotype: Normal (carrier) |
| Lowercase a | Uppercase A lower case a Phenotype: Normal (carrier) | a a both lowercase Phenotype: albino |

The square is as follows:

| Gametes | Uppercase D | Lowercase d |
|---|---|---|
| Lowercase d | Uppercase D lowercase d Phenotype: Dwarf | d d both lowercase Phenotype: Normal |
| Lowercase d | Uppercase D lowercase d Phenotype: Dwarf | d d both lowercase Phenotype: Normal |

**Further Reading** R. A. Sturm, A golden age of human pigmentation genetics, *Trends in Genetics* 22:464–468 (2006).

**INTERPRET THE DATA**

- A histogram is a bar graph that shows the distribution of numeric data (here, the number of dark-skin alleles). To make a histogram of the allele distribution, put skin color (as the number of dark-skin alleles) along the x-axis and predicted number of offspring (out of 64) with each phenotype on the y-axis. There are no gaps in these allele data, so draw the bars next to each other with no space in between.
- You can see that the skin-color phenotypes are not distributed uniformly.
- Which phenotype has the highest frequency? Draw a vertical dashed line through that bar.
- Distributions of values like this one tend to show one of several common patterns. Sketch a rough curve that approximates the values and look at its shape. Is it symmetrically distributed around a central peak value (a “normal distribution,” sometimes called a bell curve); is it skewed to one end of the x-axis or the other (a “skewed distribution”); or does it show two apparent groups of frequencies (a “bimodal distribution”)? Explain the reason for the curve’s shape. (It will help to read the text description that supports Figure 14.13.)

**Instructors:** A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

## Nature and Nurture: The Environmental Impact on Phenotype

Another departure from simple Mendelian genetics arises when the phenotype for a character depends on environment as well as genotype. A single tree, locked into its inherited genotype, has leaves that vary in size, shape, and greenness, depending on their exposure to wind and sun. For humans, nutrition influences height, exercise alters build, sun-tanning darkens the skin, and experience improves performance on intelligence tests. Even identical twins, who are genetic equals, accumulate phenotypic differences as a result of their unique experiences.

Whether human characters are more influenced by genes or the environment—in everyday terms, nature versus nurture—is a debate that we will not attempt to settle here. We can say, however, that a genotype generally is not associated with a rigidly defined phenotype, but rather with a range of phenotypic possibilities due to environmental influences **(Figure 14.14)**. For some characters, such as the ABO blood group system, the phenotypic range is extremely narrow; that is, a given genotype mandates a very specific phenotype. Other characters, such as a person's blood count of red and white cells, vary quite a bit, depending on such factors as the altitude, the customary level of physical activity, and the presence of infectious agents.

**Figure 14.14**

```
The effect of environment on phenotype.
```

Two photos. Hydrangeas grown in basic soil are bright pink in figure a. Hydrangeas grown in acidic soil with free aluminum are bluish-violet in figure b.

*The outcome of a genotype lies within a phenotypic range that depends on the environment in which the genotype is expressed. For example, the acidity and free aluminum content of the soil affect the color of hydrangea flowers, which range from pink (basic soil) to blue-violet (acidic soil). Free aluminum is necessary for bluer colors.*

Generally, the phenotypic range is broadest for polygenic characters. Environment contributes to the quantitative nature of these characters, as we have seen in the continuous variation of skin color. Geneticists refer to such characters as **multifactorial**, meaning that many factors, both genetic and environmental, collectively influence phenotype.

**BBC Video**

Genetics vs. Environment

image

Play

image

Play

## A Mendelian View of Heredity and Variation

We have now broadened our view of Mendelian inheritance by exploring degrees of dominance as well as multiple alleles, pleiotropy, epistasis, polygenic inheritance, and the phenotypic impact of the environment. How can we integrate these refinements into a comprehensive theory of Mendelian genetics? The key is to make the transition from the reductionist emphasis on single genes and phenotypic characters to the emergent properties of the organism as a whole, one of the themes of this book.

The term *phenotype* can refer not only to specific characters, such as flower color and blood group, but also to an organism in its entirety—*all* aspects of its physical appearance, internal anatomy, physiology, and behavior. Similarly, the term *genotype* can refer to an organism's entire genetic makeup, not just its alleles for a single genetic locus. In most cases, a gene's impact on phenotype is affected by other genes and by the environment. In this integrated view of heredity and variation, an organism's phenotype reflects its overall genotype and unique environmental history.

Considering all that can occur in the pathway from genotype to phenotype, it is indeed impressive that Mendel could uncover the fundamental principles governing the transmission of individual genes from parents to offspring. Mendel's laws of segregation and of independent assortment explain heritable variations in terms of alternative forms of genes (hereditary "particles," now known as the alleles of genes) that are passed along, generation after generation, according to simple rules of probability. This theory of inheritance is equally valid for peas, flies, fishes, birds, and human beings—indeed, for any organism with a sexual life cycle. Furthermore, by extending the principles of segregation and independent assortment to help explain such hereditary patterns as epistasis and quantitative characters, we begin to see how broadly Mendelian genetics applies. From Mendel's abbey garden came a theory of inheritance that anchors modern genetics. In the last section of this chapter, we will apply Mendelian genetics to human inheritance, with emphasis on the transmission of hereditary diseases.

## Concept Check 14.3

- *Incomplete dominance* and *epistasis* are both terms that define genetic relationships. What is the most basic distinction between these terms?
- If a man with type AB blood marries a woman with type O, what blood types would you expect in their children? What fraction would you expect of each type?

- **WHAT IF?** A rooster with gray feathers and a hen of the same phenotype produce 15 gray, 6 black, and 8 white chicks. What is the simplest explanation for the inheritance of these colors in chickens? What phenotypes would you expect in the offspring of a cross between a gray rooster and a black hen?

*For suggested answers, see*Appendix A.**

---

# Concept 14.4: Many human traits follow Mendelian patterns of inheritance

The study of human genetics is fueled by our desire to understand our own inheritance and to develop treatments for genetically based diseases. While peas are convenient subjects for genetic research, humans are not. The human generation span is long (about 20 years), human parents produce many fewer offspring than peas, and breeding experiments would be unethical!

## Pedigree Analysis

In place of breeding experiments, geneticists analyze the results of human matings that have already occurred. They collect information about a family's history for a particular trait and assemble this information into a family tree describing the trait across the generations—a family **pedigree**.

**Figure 14.15a** shows a three-generation pedigree that traces the occurrence of a pointed contour of the hairline on the forehead. We simplify the genetics of this trait, called a widow's peak, by assuming it is due to a dominant allele, W. (In reality, other genes are probably also involved.) In our model, because the widow's peak allele is dominant, all individuals who lack a widow's peak must be homozygous recessive (ww). The two grandparents with widow's peaks must have the Ww genotype since some of their offspring are homozygous recessive. The offspring in the second generation who *do* have widow's peaks must also be heterozygous because they are the products of Ww×ww matings. The third generation in this pedigree consists of two sisters. The one who has a widow's peak could be either homozygous (WW) or heterozygous (Ww), given what we know about the genotypes of her parents (both Ww).

**Figure 14.15**

```
    Pedigree analysis.
```

Pedigree showing three generations of a family with respect to widow's peak and ability to taste PTC. Images of two sisters shown in the third generation; one with widow's peak and one without.

*Each pedigree traces a trait through three generations of the same family; the two traits have different inheritance patterns. For the sake of understanding genetic principles, each trait is shown as being determined by two alleles of a single gene—a simplification, since other genes likely also affect these two characters.*

Part a.

- First generation (grandparents): Pedigree drawn with symbols as follows from left to right: First grandparents: Shaded Square (uppercase W lowercase w) with a line drawn to circle (w w, both lowercase). Space and then second grandparents: square (w w both lowercase) with a line drawn to shaded circle (uppercase W lower case w).
- Second generation (parents, uncles, and aunts): Children of first grandparents: Shaded circle (uppercase W lowercase w), square (w w both lowercase), square (w w both lowercase), shaded square (uppercase W lowercase w). This individual has a line drawn to the first child of the second grandparents, a shaded circle (uppercase W lowercase w). Next is a circle (w w both lowercase).
- Third generation: children of the couple in the second generation: Shaded circle (W W both uppercase or uppercase W lowercase w). This individual is shown next to a photograph of a woman with Widow's peak. Next is a circle (w w both lowercase). This individual is shown next to a photograph of a woman with no widow's peak.
- Is a widow's peak a dominant or recessive trait? Tips for pedigree analysis: Notice in the third generation that the second-born daughter lacks a widow's peak, although both of her parents had the trait. Such a pattern indicates that the trait is due to a dominant allele. If it were due to a recessive allele, and both parents had the recessive phenotype (straight hairline), all of their offspring would also have the recessive phenotype.

Part b.

- Pedigree tracing the ability to taste P T C through 3 generations of a family. In the third generation, photos show two sisters; one can taste PTC and the other can't.
- First generation (grandparents): Pedigree drawn with symbols as follows from left to right: first grandparents: Square (uppercase T lowercase t) with line drawn to circle (uppercase T lowercase t). Space then second set of grandparents: shaded square (tt both lowercase) with line drawn to circle (uppercase T lowercase t).
- Second generation (parents, aunts, uncles). Children of first set of grandparents: Circle (T T both uppercase or uppercase T lowercase t), shaded square (t t both lowercase), shaded square (t t both lowercase), square (uppercase T lowercase t). This individual has a line drawn to child of second

grandparents: circle (uppercase T lowercase t) and the other child of the second grandparents is a shaded circle (t t both lowercase).

- Children of the second-generation couple are in the third generation (two sisters). First a shaded circle (t t both lowercase, next to a photo of one sister who is smiling because she cannot taste P T C. Then a circle (TT both uppercase or uppercase T lowercase t, next to a photo of her sister who is grimacing because she can taste the bitter chemical P T C.

**Figure 14.15b** is a pedigree of the same family, but this time we focus on a recessive trait, the inability of individuals to taste a chemical called PTC (phenylthiocarbamide). Compounds similar to PTC are found in broccoli, brussels sprouts, and related vegetables and account for the bitter taste some people report when eating these foods. We'll use t for the recessive allele and T for the dominant allele, which results in the ability to taste PTC. As you work your way through the pedigree, notice once again that you can apply what you have learned about Mendelian inheritance to understand the genotypes shown for the family members.

An important application of a pedigree is to help us calculate the probability that a future child will have a particular genotype and phenotype. Suppose that the couple represented in the second generation of Figure 14.15 decides to have one more child. What is the probability that the child will have a widow's peak? This is equivalent to a Mendelian F1 monohybrid cross Ww×Ww, and therefore the probability that a child will inherit a dominant allele and have a widow's peak is 34(14WW+12Ww). What is the probability that the child will be unable to taste PTC? We can also treat this as a monohybrid cross Tt×Tt, but this time we want to know the chance that the offspring will be homozygous recessive (tt). That probability is 14. Finally, what is the chance that the child will have a widow's peak *and* be unable to taste PTC? Assuming that the genes for these two characters are on different chromosomes, the two pairs of alleles will assort independently in this dihybrid cross WwTt×WwTt. Therefore, we can use the multiplication rule:

34(chance of widow's peak)×14(chance of inability to taste PTC)=316
(chance of widow's peak and inability to taste PTC).

Pedigrees are a more serious matter when the alleles in question cause disabling or deadly diseases instead of innocuous human variations such as hairline or inability to taste a harmless chemical. However, for disorders inherited as simple Mendelian traits, the same techniques of pedigree analysis apply.

## Recessively Inherited Disorders

Thousands of genetic disorders are known to be inherited as simple recessive traits. These disorders range in severity from relatively mild, such as albinism (lack of pigmentation, which results in susceptibility to skin cancers and vision problems), to life-threatening, such as cystic fibrosis.

### The Behavior of Recessive Alleles

How can we account for the behavior of alleles that cause recessively inherited disorders? Recall that genes code for proteins of specific function. An allele that causes a genetic disorder (let's call it allele a) codes for either a malfunctioning protein or no protein at all. In the case of disorders classified as recessive, heterozygotes (Aa) typically have the normal phenotype because one copy of the normal allele (A) produces a sufficient amount of the specific protein. Thus, a recessively inherited disorder shows up only in the homozygous individuals (aa) who inherit a recessive allele from each parent. Although phenotypically normal with regard to the disorder, heterozygotes may transmit the recessive allele to their offspring and thus are called **carriers**. **Figure 14.16** illustrates these ideas using albinism as an example.

**Figure 14.16**

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Albinism: a recessive trait.
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Image of two sisters, one with albinism and one with normal coloration. Punnett square showing the results of cross between parents with normal phenotypes who are both carriers for albinism.

*One of the two sisters shown here does not have albinism; the other does. Most recessive homozygotes are born to parents who are carriers of the disorder but themselves have a normal phenotype, the case shown in the Punnett square.*

Parents both normal phenotypes, and both with genotypes uppercase A lowercase a.

Probabilities for phenotypes in their children are one-fourth normal, one-half normal carriers, and one-fourth albino.

| Gametes | Uppercase A | Lowercase a |
|---|---|---|
| Uppercase A | A A both uppercase Phenotype: normal | Uppercase A lower case a Phenotype: Normal (carrier) |
| Lowercase a | Uppercase A lower case a Phenotype: Normal (carrier) | a a both lowercase Phenotype: albino |

*What is the probability that the sister without albinism is a carrier of the albinism allele?*

*For suggested answer, see*Appendix A.**

Most people who have recessive disorders are born to parents who are carriers of the disorder but have a normal phenotype, as is the case shown in the Punnett square in Figure 14.16. A mating between two carriers corresponds to a Mendelian F1 monohybrid cross, so the predicted genotypic ratio for the offspring is 1AA:2

Aa:1aa. Thus, each child has a 14 chance of inheriting a double dose of the recessive allele; in the case of albinism, such a child will have albinism. From the genotypic ratio, we also can see that out of three offspring with the *normal* phenotype (one AA plus two Aa), two are predicted to be heterozygous carriers, a 23 chance. Recessive homozygotes could also result from Aa×aa and aa×aa matings, but if the disorder is lethal before reproductive age or results in sterility (neither of which is true for albinism), no aa individuals will reproduce. Even if recessive homozygotes are able to reproduce, this will occur relatively rarely because such individuals are much less common in the population than heterozygous carriers (for reasons we'll examine in Concept 23.2).

In general, genetic disorders are not evenly distributed among all groups of people. For example, the incidence of Tay-Sachs disease, which we described earlier in this chapter, is disproportionately high among Ashkenazic Jews, Jewish people whose ancestors lived in central Europe. In that population, Tay-Sachs disease occurs in one out of 3,600 births, an incidence about 100 times greater than that among non-Jews or Mediterranean (Sephardic) Jews. This uneven distribution results from the different genetic histories of the world's peoples during less technological times, when populations were more geographically (and hence genetically) isolated.

When a disease-causing recessive allele is rare, it is relatively unlikely that two carriers of the same harmful allele will meet and mate. The probability of passing on recessive traits increases greatly, however, if the man and woman are close relatives (for example, siblings or first cousins). This is because people with recent common ancestors are more likely to carry the same recessive alleles than are unrelated people. Thus, these consanguineous ("same blood") matings, indicated in pedigrees by double lines, are more likely to produce offspring homozygous for recessive traits—including harmful ones. Such effects can be observed in many types of domesticated and zoo animals that have become inbred.

Although geneticists generally agree that inbreeding causes an increase in autosomal recessive conditions compared to those resulting from matings between unrelated parents, they debate exactly how much human consanguinity increases the risk of inherited diseases. For one thing, many harmful alleles have such severe effects that a homozygous embryo spontaneously aborts long before birth. Most societies and cultures have laws or taboos forbidding marriages between close relatives, some for social or economic reasons. These rules may have also resulted from the empirical observation that in most populations, stillbirths and physical or biochemical abnormalities at birth are more common when parents are closely related.

## Cystic Fibrosis

The most common lethal genetic disease in the United States is **cystic fibrosis**, which strikes one out of every 2,500 people of European descent but is much rarer in other groups. Among people of European descent, one out of 25 4% are carriers of the cystic fibrosis allele. The normal allele for this gene codes for a membrane protein that functions in the transport of chloride ions between certain cells and the extracellular fluid. These chloride transport channels are defective or absent in the plasma membranes of children who

inherit two recessive alleles for cystic fibrosis. The result is an abnormally high concentration of intracellular chloride, which causes an uptake of water due to osmosis. This in turn causes the mucus that coats certain cells to become thicker and stickier than normal and to build up in the pancreas, lungs, digestive tract, and other organs. Multiple (pleiotropic) effects result, including poor absorption of nutrients from the intestines, chronic bronchitis, and recurrent bacterial infections.

Untreated, cystic fibrosis can cause death by the age of 5. Daily doses of antibiotics to stop infection, gentle pounding on the chest to clear mucus from clogged airways, and other therapies can prolong life. In the United States, more than half of those with cystic fibrosis now survive into their 40s.

## Sickle-Cell Disease: A Genetic Disorder with Evolutionary Implications

**EVOLUTION** The most common inherited disorder among people of African descent is **sickle-cell disease**, which affects one out of 400 African-Americans. Sickle-cell disease is caused by the substitution of a single amino acid in the hemoglobin protein of red blood cells; in homozygous individuals, all hemoglobin is of the sickle-cell (abnormal) variety. When the oxygen content of an affected individual's blood is low (at high altitudes or under physical stress, for instance), the sickle-cell hemoglobin proteins aggregate into long fibers that deform the red cells into a sickle shape (see Figure 5.19). Sickled cells may clump and clog small blood vessels, often leading to other symptoms throughout the body, including physical weakness, pain, organ damage, and even stroke and paralysis. Regular blood transfusions can ward off brain damage in children with sickle-cell disease, and new drugs can help prevent or treat other problems. There is currently no widely available cure, but the disease is the target of ongoing gene therapy research.

Although two sickle-cell alleles are necessary for an individual to have sickle-cell disease and thus the condition is considered recessive, the presence of one sickle-cell allele can affect the phenotype. Thus, at the organismal level, the normal allele is incompletely dominant to the sickle-cell allele **(Figure 14.17)**. At the molecular level, the two alleles are codominant; both normal and abnormal (sickle-cell) hemoglobins are made in heterozygotes (carriers), who are said to have *sickle-cell trait*. (Here the word "trait" is used to distinguish this condition from sickle-cell disease, not as it was defined earlier—as any variant of a character.) Heterozygotes are usually healthy but may suffer some symptoms during long periods of reduced blood oxygen.

**Figure 14.17**

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Sickle-cell disease and sickle-cell trait.
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Sickle-cell disease and sickle-cell trait.

Part a. Homozygote with sickle-cell disease. Weakness, anemia, pain and fever, organ damage. Sickle cell alleles on two homologous chromosomes flows to sickle-cell hemoglobin proteins which goes with low O 2 to part of a fiber of sickle-cell hemoglobin proteins. Long fibers cause red blood cells to be sickle-shaped, leading to sickle cell disease.

Part b. Heterozygote with sickle-cell trait. Some symptoms when blood oxygen is very low; reduction of malaria symptoms. Pair of homologous chromosomes, one with a sickle-cell allele and one with a normal allele, flows to both sickle cell and normal hemoglobin proteins, flows with low O 2 to part of a sickle-cell fiber and normal hemoglobin proteins, which flows to both sickled and normal red blood cells, which flows to sickle-cell trait.

### HHMI BioInteractive Animation

Sickle-Cell Disease

image

Play

image

Play

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Considering the severe detrimental effects of this allele in homozygotes, there is an unusually high percentage of heterozygotes. For example, about one out of ten African-Americans have sickle-cell trait. Why haven't evolutionary processes resulted in the disappearance of the allele among this population? One explanation is that having a single copy of the sickle-cell allele reduces the frequency and severity of malaria attacks, especially among young children. The malaria parasite spends part of its life cycle in red blood cells (see Figure 28.18), and the presence of even heterozygous amounts of sickle-cell hemoglobin results in lower parasite densities and hence reduced malaria symptoms. Thus, in tropical Africa, where infection with the malaria parasite is common, the sickle-cell allele confers an advantage to heterozygotes even though it is harmful in the homozygous state. (The balance between these two effects will be discussed in Concept 23.4; see Make Connections Figure 23.18.) The relatively high frequency of African-Americans with sickle-cell trait is a vestige of their tropical African ancestry.

### HHMI BioInteractive Video

The Making of the Fittest: Natural Selection in Humans (Sickle-Cell Disease)

image

Play

image

Play

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## Dominantly Inherited Disorders

Although many harmful alleles are recessive, some disorders are due to dominant alleles. One example is *achondroplasia*, a form of dwarfism that occurs in one of every 25,000 people. Heterozygous individuals have the dwarf phenotype **(Figure 14.18)**. Therefore, all people who do not have achondroplasia—99.99% of the population—are homozygous recessive. Like the presence of extra fingers or toes mentioned earlier, achondroplasia is a trait for which the recessive allele is much more prevalent than the corresponding dominant allele.

**Figure 14.18**

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Achondroplasia: a dominant trait.
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Punnett square shows the outcome of cross between a heterozygous dwarf and a normal (homozygous recessive) parent. Half their children will be heterozygous (dwarves) and half are normal (homozygous recessive).

See table next tab

Dr. Michael C. Ain has achondroplasia, a form of dwarfism caused by a dominant allele. This has inspired his work: He is a specialist in the repair of bone defects caused by achondroplasia and other disorders. The dominant allele (D) might have arisen as a mutation in the egg or sperm of a parent or could have been inherited from an affected parent, as shown for an affected father in the Punnett square.

Unlike achondroplasia, which is relatively harmless, some dominant alleles cause lethal diseases. Those that do are much less common than recessive alleles that have lethal effects. A lethal recessive allele is only lethal when homozygous; it can be passed from one generation to the next by heterozygous carriers because the carriers themselves have normal phenotypes. A lethal dominant allele, however, often causes the death of afflicted individuals before they can mature and reproduce, and in this case the allele is not passed on to future generations.

A lethal dominant allele may be passed on, though, if the lethal disease symptoms first appear after reproductive age. In these cases, the individual may already have transmitted the allele to his or her children. For example, a degenerative disease of the nervous system, called **Huntington's disease**, is caused by a lethal dominant allele that has no obvious phenotypic effect until the individual is about 35 to 45 years old. Once the deterioration of the nervous system begins, it is irreversible and inevitably fatal. As with other dominant traits, a child born to a parent with the Huntington's disease allele has a 50% chance of inheriting the allele and the disorder (see the Punnett square in Figure 14.18). In the United States, this disease afflicts about one in 10,000 people.

At one time, the onset of symptoms was the only way to know if a person had inherited the Huntington's allele, but this is no longer the case. By analyzing DNA samples from a large family with a high incidence of the disorder, geneticists tracked the Huntington's allele to a locus near the tip of chromosome 4, and the gene was sequenced in 1993. This information led to the development of a test that could detect the presence of the Huntington's allele in an individual's genome. (The methods that make such tests possible are discussed in Concepts 20.1 and 20.4.) The availability of this test poses an agonizing dilemma for those with a family history of Huntington's disease. Some individuals may want to be tested for this disease, whereas others may decide it would be too stressful to find out.

## Interview

**Interview with Nancy Wexler: Mapping the gene that causes Huntington's disease**

Nancy Wexler.

## Multifactorial Disorders

The hereditary diseases we have discussed so far are sometimes described as simple Mendelian disorders because they result from abnormality of one or both alleles at a single genetic locus. Many more people are susceptible to diseases that have a multifactorial basis—a genetic component plus a significant environmental influence. Heart disease, diabetes, cancer, alcoholism, certain mental illnesses such as schizophrenia and bipolar disorder, and many other diseases are multifactorial. In these cases, the hereditary component is polygenic. For example, many genes affect cardiovascular health, making some of us more prone than others to heart attacks and strokes. No matter what our genotype, however, our lifestyle has a tremendous effect on phenotype for cardiovascular health and other multifactorial characters. Exercise, a healthful diet, abstinence from smoking, and an ability to handle stressful situations all reduce our risk of heart disease and some types of cancer.

## Genetic Testing and Counseling

Avoiding simple Mendelian disorders is possible when the risk of a particular genetic disorder can be assessed before a child is conceived or during the early stages of the pregnancy. Many hospitals have genetic counselors who can provide information to prospective parents concerned about a family history for a specific disease. Fetal and newborn testing can also reveal genetic disorders.

## Counseling Based on Mendelian Genetics and Probability Rules

Consider the case of a hypothetical couple, Tyler and Lily. Each had a brother who died from the same recessively inherited lethal disease. Before conceiving their first child, Tyler and Lily seek genetic counseling to determine the risk of having a child with the disease. From the information about their brothers, we know that both parents of Tyler and both parents of Lily must have been carriers of the recessive allele. Thus, Tyler and Lily are both products of Aa×Aa crosses, where a symbolizes the allele that causes this particular disease. We also know that Tyler and Lily are not homozygous recessive (aa) because they do not have the disease. Therefore, their genotypes are either AA or Aa.

Given a genotypic ratio of 1AA:2Aa:1aa for offspring of an Aa×Aa cross, Tyler and Lily each have a 23 chance of being carriers (Aa). According to the rule of multiplication, the overall probability of their firstborn having the disorder is 23 (the chance that Tyler is a carrier) times 23 (the chance that Lily is a carrier) times 14 (the chance of two carriers having a child with the disease), which equals 19. Suppose that Lily and Tyler decide to have a child—after all, there is an 89 chance that their baby will not have the disorder. If, despite these odds, their child is born with the disease, then we would know that *both* Tyler and Lily are, in fact, carriers (Aa genotype). If both Tyler and Lily are carriers, there is a 14 chance that any subsequent child this couple has will have the disease. The probability is higher for subsequent children because the diagnosis of the disease in the first child established that both parents are carriers, not because the genotype of the first child affects in any way that of future children.

When we use Mendel's laws to predict possible outcomes of matings, it is important to remember that each child represents an independent event in the sense that its genotype is unaffected by the genotypes of older siblings. Suppose that Tyler and Lily have three more children, and *all three* have the hypothetical hereditary disease. There is only one chance in 64(14×14×14) that such an outcome will occur. Despite these circumstances, the chance that a fourth child of this couple will have the disease remains 14.

## Tests for Identifying Carriers

Most children with recessive disorders are born to parents with normal phenotypes. The key to accurately assessing the genetic risk for a particular disease is therefore to find out whether the prospective parents are heterozygous carriers of the recessive allele. Tests are available that can distinguish individuals of normal phenotype who are dominant homozygotes from those who are heterozygous carriers, using blood or cells from the inside of the cheek. Hundreds of different recessive alleles can be detected, including those for Tay-Sachs disease, sickle-cell disease, and cystic fibrosis.

These tests for identifying carriers enable people with family histories of genetic disorders to make informed decisions about having children, including whether to do genetic testing of the fetus, should they decide to become pregnant. The results can also lead carriers to alert family members that they may also be carriers and might choose to be tested. The tests raise other issues as well: Could carriers be denied health or life insurance or lose the jobs providing those benefits, even though they themselves are healthy? The Genetic Information Nondiscrimination Act, in place since 2008, allays these concerns by prohibiting discrimination in employment or health insurance coverage based on genetic test results. This is an important act that protects patients' rights but is unknown to many, including physicians. Genetic counselors, trained in a growing number of professional programs, can help individuals understand their genetic test results. Once test results are clearly understood, those with certain results may face difficult decisions. Advances in biotechnology offer the potential to reduce human suffering, but along with them come ethical issues that require conscientious deliberation.

## Fetal Testing

Two types of testing can be done to look for genetic disorders in the fetus: screening tests and diagnostic tests. Screening tests are generally noninvasive and can be followed by diagnostic tests if screening results suggest that a genetic disorder might be present or if there are other risks, as when both parents are carriers.

Screening tests include imaging and blood tests. Imaging techniques allow a physician to examine a fetus directly for major anatomical abnormalities that might not show up in genetic tests. In the *ultrasound* technique, for example, reflected sound waves are used to produce an image of the fetus by a simple noninvasive procedure. An ultrasound is often accompanied by a blood test that looks for fetal proteins that might signal a genetic disorder.

Medical scientists have also developed methods for isolating and analyzing fetal DNA that has escaped into the maternal blood (*cell-free fetal DNA*), using various techniques. Cell-free fetal DNA tests and other blood tests are increasingly being used as noninvasive prenatal screening tests for certain chromosomal defects and disorders; a positive result indicates to the parents that further diagnostic testing should be considered.

One diagnostic test that can determine whether the developing fetus has a serious recessive disease is **amniocentesis**, which can be performed starting at the 15th or 16th week of pregnancy **(Figure 14.19a)**. In this procedure, a physician inserts a needle into the uterus and extracts about 10 mL of amniotic fluid, the liquid that bathes the fetus. Some genetic disorders can be detected from the presence of certain proteins or hormones in the amniotic fluid itself. Tests for thousands of other disorders (such as cystic fibrosis) are performed on DNA from fetal cells sloughed off into the amniotic fluid using targeted DNA sequencing or a technique called a *DNA microarray* (see Figure 20.13) that tests for many alleles at the same time. A karyotype of these cultured cells can also confirm certain chromosomal defects that may not have been suggested by cell-free fetal DNA tests.

**Figure 14.19**

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Diagnostic fetal testing for genetic disorders.
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Diagnostic fetal testing for genetic disorders.

*Most testing for genetic abnormalities is now DNA-based. In some cases biochemical tests are used to detect particular disorders, and karyotyping may be done to see whether the chromosomes of the fetus are normal in number and appearance.*

Part a. Amniocentesis summarized in three steps.

- 1. A sample of amniotic fluid can be taken starting at the fifteenth week of pregnancy. Drawing of fetus in utero with placenta, fetus, uterus, cervix labeled. An ultrasound monitor is in place while the amniotic fluid is withdrawn using a needle.
- 2. Biochemical and genetic tests can be performed immediately on the amniotic fluid or later on the cultured cells. Arrows flow from withdrawn fluid in needle to fluid and fetal cells in a test tube, where after several hours biochemical and genetics tests are run. 3. Fetal cells must be cultured for several weeks to obtain sufficient numbers for karyotyping. Arrows flow from fetal cells in test tube to petri dish, where fetal cells are cultured for several weeks, after which a karyotype is made. Image of karyotype shows the fetus' 23 pairs of homologous chromosomes.

Part b. Chorionic villus sampling (CVS) described in two steps.

- 1. A sample of chorionic villus tissue can be taken as early as the tenth week of pregnancy. Drawing of fetus in utero with placenta, fetus, Chorionic villi, uterus and cervix labeled. Drawing shows suction tube inserted through cervix to collect fetal cells.
- 2. Karyotyping and biochemical and genetic tests can be performed on the fetal cells immediately, providing results within a day or so. Arrows flow to fetal cells in test tube. After several hours, biochemical and genetic tests can be run, and a karyotype produced. Image of karyotype shows the fetus' 23 pairs of homologous chromosomes.

In an alternative diagnostic technique that is called **chorionic villus sampling (CVS)**, a physician inserts a narrow tube through the cervix into the uterus and suctions out a tiny sample of tissue from the placenta, the organ that transmits nutrients and fetal wastes between the fetus and the female parent **(Figure 14.19b)**. The cells of the chorionic villi of the placenta—the portion sampled—are derived from the fetus and have the same genotype and DNA sequence as the new individual. DNA from these cells can be analyzed for

the presence of many genetic disorders. The main advantage of CVS is that it can be performed as early as the 10th or 11th week of pregnancy.

Ultrasound and isolation of fetal cells or DNA from maternal blood pose no known risk to either the female parent or the fetus, while amniocentesis and CVS can cause complications in a small percentage of cases (for both, between 1 in 1,000 and 1 in 500 cases). Amniocentesis or CVS for diagnostic testing is generally offered to females over age 35, due to their increased risk of bearing a child with Down syndrome or other chromosomal disorders, and may also be offered to younger females if there are risk factors or conditions suggested by family history. If the fetal tests reveal a serious disorder, the parents may face the difficult choice of either terminating the pregnancy or preparing to care for a child with a genetic disorder, one that might even be fatal. Parental and fetal screening for Tay-Sachs alleles done since 1980 has reduced the number of children born with this incurable disease by 90%.

### Newborn Screening

Some genetic diseases and disorders can be detected at birth by simple biochemical tests that are routinely performed on blood from the baby's heel in most hospitals in the United States. One common screening program is for phenylketonuria (PKU), a recessively inherited disorder that occurs in about one out of every 10,000–15,000 births in the United States. Children with this disease cannot properly metabolize the amino acid phenylalanine. This compound and its by-product, phenylpyruvate, can accumulate to toxic levels in the blood, causing severe intellectual disability. However, if PKU is detected in the newborn, a specialized diet low in phenylalanine will usually allow for typical development. (Among many other substances, this diet excludes the artificial sweetener aspartame, which contains phenylalanine.)

The number of newborn genetic tests, many for metabolic disorders or enzyme deficiencies, is over 100 and ever-increasing; each state regulates which tests are used. If screening identifies a baby with a disease or disorder for which treatment can be started early on, that individual has improved chances for better life quality. Some diseases, like PKU, are treated by dietary changes and others by replacement of missing enzymes. Unfortunately, there are quite a few diseases for which no treatment is yet known.

Fetal and newborn screening for serious inherited diseases, tests for identifying carriers, and genetic counseling all rely on the Mendelian model of inheritance. We owe the "gene idea"—the concept of heritable factors transmitted according to simple rules of chance—to the elegant quantitative experiments of Gregor Mendel. The importance of his discoveries was overlooked by most biologists until the early 1900s, decades after he reported his findings. In the next chapter, you will learn how Mendel's laws have their physical basis in the behavior of chromosomes during sexual life cycles and how the synthesis of Mendelian genetics and a chromosome theory of inheritance catalyzed progress in genetics.

## Concept Check 14.4

- Lucia and Jared each have a sibling with cystic fibrosis, but neither Lucia nor Jared nor any of their parents have the disease. Calculate the probability that if this couple has a child, the child will have cystic fibrosis. What would be the probability if a test revealed that Jared is a carrier but Lucia is not? Explain your answers.
- **MAKE CONNECTIONS** Explain how the change of a single amino acid in hemoglobin leads to the aggregation of hemoglobin into long fibers. (Review Figures 5.14, 5.18, and 5.19.)
- Juanita was born with six toes on each foot, a dominant trait called polydactyly. Two of her five siblings and her mother, but not her father, also have extra digits. What is Juanita's genotype for the number-of-digits character? Explain your answer. Use D and d to symbolize the alleles for this character.
- **MAKE CONNECTIONS** In Table 14.1, note the phenotypic ratio of the dominant to recessive trait in the F2 generation for the monohybrid cross involving flower color. Then determine the phenotypic ratio for the offspring of the second-generation couple in Figure 14.15b. What accounts for the difference in the two ratios?

*For suggested answers, see*Appendix A.**

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